

Resonant Gamma Jets and Directional Zeta Identities

Discrete Stieltjes products, shifted polylogarithms,
and local formulas for higher spectral derivatives

ProveIt research continuation
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Abstract

We develop six compatible exact calculi beyond the current polylogarithm manuscript. A discrete product identity evaluates convergent sums containing arbitrarily many generalized Stieltjes functions, with an explicit finite remainder. A holomorphic remainder for the double Hurwitz zeta function separates its two intersecting polar divisors, yielding the complete dependence of its constant term on a transverse direction or a curved approach. Symmetrized nested sums admit an all-depth, all-index partition formula. The higher spectral derivatives of a renormalized reciprocal-Gamma product at every moving Gamma zero are then reduced to a finite Bell algorithm and a finite triangular family of ordinary antiderivatives. An independent shifted-gap identity gives complete reductions of complementary-shift double polylogarithms throughout the open argument disk. Finally, higher derivatives of polynomially weighted spectral zeta functions satisfy finite-difference identities with explicitly local right sides; a six-term second-derivative identity has a Vandermonde evaluation. It belongs to an alternating Vandermonde family with any number of factors, including exact identities for fifth and higher spectral derivatives. All stated identities are proved, with domains and normalizations. The classical symmetric-sum, parity, and multiplicative-anomaly mechanisms are credited explicitly. Numerical checks are reproducible diagnostics, not substitutes for proofs. The short Gaussian S_6 and S_8 evaluations and arithmetic minimality questions remain open.

Source baseline. Repository revision `16c7e342d7a4a15f59911f322eb90b7f96c2a8b5`; the canonical manuscript and all five archives present in `docs/incoming` at that revision.

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1 Scope, provenance, and conventions

The immediate source is the collective manuscript *Polylogarithms and their Arithmetic Bridges*, together with the five incoming archives in the pinned revision [1, 2, 3, 4, 5, 6]. The manuscript already contains a substantial theory of periodic Stieltjes correlations, distributional derivatives, dilation contacts, and log-Gamma convolutions. Repeating those results would not answer the request for a research continuation. Here the focus is on discrete translation, independent spectral directions, moving zeros, and nonperiodic shifted sums.

The incoming *Nested Harmonic Jets* report asks, among other things, for non-diagonal spectral directions, path-dependent multiple Laurent finite parts, and an explicit algorithm for higher spectral values at moving Gamma zeros. We answer the algorithmic part of the last question, give a complete transverse-path calculation at depth two, and give every Laurent coefficient of the fully symmetrized family at arbitrary depth. The unsymmetrized higher-depth problem and minimality of the resulting constant classes are not settled by these results.

1.1 What is established here

Topic	Exact conclusion	Location
Discrete Stieltjes products	One explicit correction term evaluates the grouped sum for every number of factors and all Laurent indices; the finite-cutoff remainder is exact.	Section 2
Intersecting nested poles	A holomorphic two-variable remainder, all its Taylor coefficients, straight-direction finite parts, and the cubic-jet correction for curved paths.	Section 3
Symmetrized mixed jets	A finite partition formula for every Laurent coefficient in every depth, including an explicit depth-three constant term.	Section 4
Moving Gamma zeros	A Bell-polynomial formula for every derivative, a closed second derivative, and a finite antiderivative construction for all necessary coordinates.	Section 5
Shifted double polylogarithms	A disk identity in two independent shifts; complete reductions at complementary shifts, including rational-shift specializations.	Section 6
Higher spectral products	Polynomiality in normalized exponents, local finite differences, alternating Vandermonde evaluations for any number of factors, and logarithmic multiplicities.	Section 7

These are proved extensions developed for this continuation. This wording does not assert historical priority for every displayed specialization. The symmetric-sum theorem is classical [8]; Laurent expansions of multiple zeta functions have an established literature [9, 10]; broad shifted cyclotomic symmetry is treated in [12, 13]; and first-derivative multiplicative anomalies are classical in substance [14, 15, 16]. The contributions should be assessed as explicit formulas, proofs, normalizations, and extensions of the repository's available results, with those antecedents retained.

1.2 Basic notation

For $a > 0$, let

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \gamma_j(a) u^j, \quad g_a(u) = \zeta(1+u, a) - \frac{1}{u}. \quad (1.1)$$

The function g_a is entire in u . Thus $\gamma_0(a) = -\psi(a)$, where $\psi = \Gamma'/\Gamma$, and $\gamma_j = \gamma_j(1)$. Spectral derivatives of Hurwitz zeta are written $\zeta^{(r)}(s, a)$; derivatives of $\gamma_j(a)$ refer to a . The standard shift and differentiation identities are [7, §25.11]

$$\gamma_j(a+1) = \gamma_j(a) - \frac{\log^j a}{a}, \quad (1.2)$$

$$\partial_a \zeta(s, a) = -s \zeta(s+1, a), \quad (1.3)$$

$$\psi^{(k)}(a) = (-1)^{k+1} k! \zeta(k+1, a) \quad (k \geq 1). \quad (1.4)$$

All logarithms of positive real numbers are real. When a complex argument is introduced, its branch is specified locally. No unspecified principal logarithm of a Gamma quotient is used.

For a meromorphic one-variable germ $f(\varepsilon)$,

$$\text{FP}_{\varepsilon=0} f = [\varepsilon^0] f. \quad (1.5)$$

The coefficient depends on the parameter ε . A function of several variables need not have a path-independent finite part. Nor is the constant Laurent coefficient automatically an algebra homomorphism. We retain every regulator until after multiplication or composition.

The exponential Bell polynomials Y_r are defined by

$$\exp \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{r \geq 0} Y_r(x_1, \dots, x_r) \frac{t^r}{r!}. \quad (1.6)$$

In particular $Y_0 = 1$, $Y_1 = x_1$, $Y_2 = x_1^2 + x_2$, and $Y_3 = x_1^3 + 3x_1x_2 + x_3$. Generalized harmonic numbers are $H_N^{(j)} = \sum_{k=1}^N k^{-j}$, with $H_0^{(j)} = 0$.

1.3 What the verification does and does not establish

The proofs below give the analytic statements. The accompanying programs check independently constructed finite coefficients, special-function values, quadrature formulas, and the signs and scale factors in the spectral identities. Exact symbolic assertions are separated from floating-point diagnostics. The package makes no claim of proof-assistant formalization or interval certification. In particular, a small residual does not establish the short S_6 or S_8 evaluations.

2 An all-degree discrete Stieltjes product identity

The elementary shift law (1.2) contains a useful exact nonlinear summation calculus. The obstacle is that a product of Stieltjes functions grows as a power of $\log x$. A single additional Stieltjes function cancels both its leading term and its first inverse-power term. This makes a telescoping identity into an ordinary convergent series.

Lemma 2.1 (The two terms needed for product cancellation). *For each fixed integer $p \geq 0$, as $x \rightarrow +\infty$,*

$$\gamma_p(x) = -\frac{\log^{p+1} x}{p+1} + \frac{\log^p x}{2x} + O\left(\frac{1 + \log^p x}{x^2}\right). \quad (2.1)$$

The expansion can be differentiated any fixed number of times in x , with the corresponding differentiated error estimate.

Proof. Use Euler–Maclaurin for $\zeta(1+u, x)$, uniformly on a small circle about $u = 0$:

$$\zeta(1+u, x) = \frac{x^{-u}}{u} + \frac{x^{-1-u}}{2} + \frac{1+u}{12}x^{-2-u} + O(x^{-4-\operatorname{Re} u}).$$

The Euler–Maclaurin remainder is holomorphic in u there after the displayed pole is removed. Its integral form permits any fixed number of u and x derivatives: each u derivative inserts a power of a logarithm, and each x derivative increases the inverse-power decay. Coefficient extraction therefore gives (2.1) with its displayed logarithmic error, and the same argument gives every fixed differentiated error estimate. This is a fixed-index use of the standard Hurwitz expansion, not an assertion uniform in growing p . $\square$

Fix $r \geq 2$ and integers $p_1, \dots, p_r \geq 0$. Put

$$K = \sum_{i=1}^r (p_i + 1), \quad A = \prod_{i=1}^r (p_i + 1), \quad c = \frac{(-1)^r K}{A}, \quad (2.2)$$

and define

$$F_{\mathbf{p}}(x) = \prod_{i=1}^r \gamma_{p_i}(x) + c\gamma_{K-1}(x). \quad (2.3)$$

The finite subset formula below is practical even when the indices p_i are unequal.

Theorem 2.2 (Arbitrary products, one correction, exact remainder). *For $a > 0$, set $x_n = n + a$ and*

$$W_{\mathbf{p}}(x) = \sum_{\emptyset \neq I \subseteq \{1, \dots, r\}} (-1)^{|I|+1} \frac{\log^{\sum_{i \in I} p_i} x}{x^{|I|}} \prod_{j \notin I} \gamma_{p_j}(x) + c \frac{\log^{K-1} x}{x}. \quad (2.4)$$

Then the grouped summand is absolutely summable, and

$$\boxed{\sum_{n=0}^{\infty} W_{\mathbf{p}}(n+a) = \prod_{i=1}^r \gamma_{p_i}(a) + c\gamma_{K-1}(a).} \quad (2.5)$$

For every integer $N \geq 0$ one has the exact finite identity

$$\sum_{n=0}^{N-1} W_{\mathbf{p}}(n+a) = F_{\mathbf{p}}(a) - F_{\mathbf{p}}(N+a). \quad (2.6)$$

In particular the remaining infinite tail is precisely $F_{\mathbf{p}}(N+a)$.

Proof. Write $h_i(x) = \log^{p_i} x/x$. The product expansion and (1.2) give

$$\prod_i \gamma_{p_i}(x) - \prod_i \gamma_{p_i}(x+1) = \sum_{\emptyset \neq I} (-1)^{|I|+1} \prod_{i \in I} h_i(x) \prod_{j \notin I} \gamma_{p_j}(x).$$

Adding c times the shift law at index $K - 1$ proves $W_{\mathbf{p}}(x) = F_{\mathbf{p}}(x) - F_{\mathbf{p}}(x + 1)$, and therefore (2.6).

For completeness, the cancellation at infinity is stronger than merely removing a divergent logarithm. Lemma 2.1 gives

$$\prod_i \gamma_{p_i}(x) = \frac{(-1)^r}{A} \log^K x + \frac{(-1)^{r-1} K \log^{K-1} x}{2A} + O\left(\frac{1 + \log^{K-1} x}{x^2}\right).$$

Both displayed terms cancel against $c\gamma_{K-1}(x)$. Thus $F_{\mathbf{p}}(x) = O((1 + \log^{K-1} x)/x^2)$, and its derivative is $O((1 + \log^{K-1} x)/x^3)$. Integrating the derivative from x to $x + 1$ proves absolute summability of the grouped $W_{\mathbf{p}}$. Passing to the limit in the exact finite identity proves (2.5). $\square$

Remark 2.3 (Grouping is part of the statement). The individual terms in (2.4) usually have divergent series. Equation (2.5) is an ordinary convergent sum of the displayed grouped expression. It does not license replacing each separate divergent series by an independently chosen finite part.

2.1 A symmetric two-index summation formula

For $m, n \geq 0$ define the absolutely convergent series

$$S_{m,n}(a) = \sum_{k=0}^{\infty} \frac{\log^m(k+a)}{k+a} \left\{ \gamma_n(k+a) + \frac{\log^{n+1}(k+a)}{n+1} - \frac{\log^n(k+a)}{2(k+a)} \right\}. \quad (2.7)$$

The braces are $O((1 + \log^n(k+a))/(k+a)^2)$.

Corollary 2.4 (All-index Stieltjes summation). *For $a > 0$ and $m, n \geq 0$,*

$$S_{m,n}(a) + S_{n,m}(a) = \gamma_m(a)\gamma_n(a) + \left(\frac{1}{m+1} + \frac{1}{n+1} \right) \gamma_{m+n+1}(a). \quad (2.8)$$

In particular,

$$S_{n,n}(a) = \frac{\gamma_n(a)^2}{2} + \frac{\gamma_{2n+1}(a)}{n+1}. \quad (2.9)$$

Proof. Apply Theorem 2.2 with $r = 2$, $p_1 = m$, and $p_2 = n$. The two copies of the last term in the braces of (2.7) combine into the equality-block correction $-\log^{m+n} x/x^2$ in (2.4). $\square$

The first cases already connect the three types of functions sought in the research programme:

$$\sum_{k=0}^{\infty} \frac{\log(k+a) - \psi(k+a) - 1/(2(k+a))}{k+a} = \frac{\psi(a)^2}{2} + \gamma_1(a), \quad (2.10)$$

$$\sum_{k=0}^{\infty} \frac{\log(k+a)}{k+a} \left\{ \gamma_1(k+a) + \frac{\log^2(k+a)}{2} - \frac{\log(k+a)}{2(k+a)} \right\} = \frac{\gamma_1(a)^2 + \gamma_3(a)}{2}. \quad (2.11)$$

The first is a useful normalization check; the second has exactly the same proof, with no numerical relation search.

2.2 Cubic and higher polygamma identities

Taking every $p_i = 0$ gives $K = r$, $A = 1$, and

$$\sum_{k=0}^{\infty} \left\{ (-\psi(x_k))^r - \left(-\psi(x_k) - \frac{1}{x_k} \right)^r + (-1)^r r \frac{\log^{r-1} x_k}{x_k} \right\} = (-\psi(a))^r + (-1)^r r \gamma_{r-1}(a). \quad (2.12)$$

For example,

$$\sum_{k=0}^{\infty} \left\{ \frac{3\psi(x_k)^2 - 3\log^2 x_k}{x_k} + \frac{3\psi(x_k)}{x_k^2} + \frac{1}{x_k^3} \right\} = -\psi(a)^3 - 3\gamma_2(a). \quad (2.13)$$

At an integer $a = N$, the recurrence $\psi(N+k) = H_{N+k-1} - \gamma$ turns this into an exact harmonic-number sum; at rational a it gives the corresponding shifted harmonic sum.

Every identity in this section can be differentiated with respect to a on compact subintervals of $(0, \infty)$. Indeed the differentiated Euler–Maclaurin estimate gives locally uniform summability of all fixed derivative orders. For reference,

$$\gamma'_n(a) = (-1)^{n+1} \left\{ \zeta^{(n)}(2, a) + n\zeta^{(n-1)}(2, a) \right\}, \quad (2.14)$$

where the second term is omitted for $n = 0$. This follows by matching coefficients in (1.3). Thus the differentiated identities remain in an explicitly stated zeta–polygamma vocabulary.

There is also an exact primitive statement. On every compact interval $[a, b] \subset (0, \infty)$,

$$\sum_{n=0}^{\infty} \int_a^b W_p(n+t) dt = \int_a^b \left\{ \prod_i \gamma_{p_i}(t) + c\gamma_{K-1}(t) \right\} dt. \quad (2.15)$$

This follows by locally uniform convergence. It fixes the integration constant through a definite interval, instead of silently assigning a normalization to an indefinite primitive.

3 Directional and curved finite parts of double Hurwitz zeta

The incoming report asks what changes when the exponent perturbations are independent, and whether its diagonal finite parts depend on the approach path. At depth two these questions have a particularly explicit answer. The meromorphic continuation itself belongs to the established multiple-zeta theory [9]. The result below identifies its full polar germ in the present normalization, then computes the resulting path dependence and all regular Taylor coefficients.

For $a > 0$ define

$$T(s, t; a) = \sum_{n>m \geq 0} (n+a)^{-s} (m+a)^{-t}. \quad (3.1)$$

The defining series is absolutely convergent when $\operatorname{Re} s > 1$ and $\operatorname{Re}(s+t) > 2$. Its order convention is essential: the outer, larger index carries s . Put

$$g_a(u) = \zeta(1+u, a) - \frac{1}{u} = \sum_{j \geq 0} g_j(a) u^j, \quad g_j(a) = \frac{(-1)^j \gamma_j(a)}{j!}.$$

Theorem 3.1 (Exact polar germ at $(1, 1)$). *There is a function $H_a(u, v)$ holomorphic near $(0, 0)$ such that*

$$\boxed{T(1+u, 1+v; a) = \frac{1}{u(u+v)} + \frac{g_a(v)}{u} + H_a(u, v).} \quad (3.2)$$

It satisfies

$$H_a(u, v) + H_a(v, u) = g_a(u)g_a(v) - \zeta(2 + u + v, a), \quad (3.3)$$

$$H_a(0, 0) = C_a := \frac{\gamma_0(a)^2 - \zeta(2, a)}{2}. \quad (3.4)$$

Thus the two local polar divisors are $u = 0$ and $u + v = 0$; in particular there is no separate polar divisor $v = 0$.

Proof. Write $x = n + a$, and use the inner finite sum to obtain, first in the absolute-convergence domain,

$$T(1 + u, 1 + v; a) = \zeta(1 + u, a)\zeta(1 + v, a) - \sum_{n \geq 0} x^{-1-u} \zeta(1 + v, x).$$

The pole-subtracted Euler–Maclaurin remainder

$$E(v, x) = \zeta(1 + v, x) - \frac{x^{-v}}{v} - \frac{1}{2}x^{-1-v}$$

is holomorphic in v near zero and is $O(x^{-2-\operatorname{Re} v})$ uniformly on compact v -sets there. Hence

$$R_a(u, v) = \sum_{n \geq 0} (n + a)^{-1-u} E(v, n + a)$$

converges normally in a neighborhood of $(0, 0)$, as do its fixed derivatives. We obtain

$$T = \zeta(1 + u, a)\zeta(1 + v, a) - \frac{\zeta(1 + u + v, a)}{v} - \frac{1}{2}\zeta(2 + u + v, a) - R_a(u, v). \quad (3.5)$$

The apparent singularity at $v = 0$ cancels. Substitution of $\zeta(1 + u, a) = u^{-1} + g_a(u)$ gives (3.2), with

$$H_a(u, v) = g_a(u)g_a(v) + \frac{g_a(u) - g_a(u + v)}{v} - \frac{1}{2}\zeta(2 + u + v, a) - R_a(u, v). \quad (3.6)$$

The divided difference is holomorphic, since its numerator vanishes at $v = 0$. Finally, the finite-sum shuffle identity continues to

$$T(s, t; a) + T(t, s; a) = \zeta(s, a)\zeta(t, a) - \zeta(s + t, a).$$

Insert (3.2) to get (3.3) and then (3.4). All steps are identities of meromorphic germs, so no choice of a one-dimensional limiting path is hidden in the proof. $\square$

3.1 Straight directions and their exact discrepancy

Here $\operatorname{FP}_{\varepsilon=0}$ means the coefficient of ε^0 in the Laurent expansion after the specified substitution. It is not a joint value at $(u, v) = (0, 0)$.

Corollary 3.2 (Every transverse straight direction). *Let $c, d \in \mathbb{C}$, with $c \neq 0$ and $c + d \neq 0$. Then*

$$T(1 + c\varepsilon, 1 + d\varepsilon; a) = \frac{1}{c(c + d)\varepsilon^2} + \frac{\gamma_0(a)}{c\varepsilon} + C_a - \frac{d}{c}\gamma_1(a) + O(\varepsilon), \quad (3.7)$$

and in particular

$$\boxed{\operatorname{FP}_{\varepsilon=0} T(1 + c\varepsilon, 1 + d\varepsilon; a) = \frac{\gamma_0(a)^2 - \zeta(2, a)}{2} - \frac{d}{c}\gamma_1(a).} \quad (3.8)$$

The case $d = 0$ is included. The difference between two admissible directions is exactly $-\gamma_1(a)(d/c - d'/c')$.

Proof. Substitute in (3.2), expand $g_a(d\varepsilon)$ through its linear term, and use (3.4). $\square$

For $c = d = 1$, this agrees with the diagonal identity $T(s, s; a) = (\zeta(s, a)^2 - \zeta(2s, a))/2$. For $d = 0$, the finite part is C_a instead. A curve contained in $u = 0$ or $u + v = 0$ cannot be evaluated by restriction of the displayed meromorphic function without an additional prescription. Curves tangent to those divisors can have higher pole orders and are outside the transversality statement.

3.2 Curvature and reparametrization also matter

The tangent direction alone does not determine the constant Laurent coefficient. The double pole reads the cubic jet of a curve.

Theorem 3.3 (Finite part along an analytic transverse curve). *Suppose*

$$u = c_1\varepsilon + c_2\varepsilon^2 + c_3\varepsilon^3 + O(\varepsilon^4), \quad v = d_1\varepsilon + d_2\varepsilon^2 + d_3\varepsilon^3 + O(\varepsilon^4),$$

where $c_1 \neq 0$ and $A := c_1 + d_1 \neq 0$. Set $B = c_2 + d_2$ and $C = c_3 + d_3$. Then

$$\begin{aligned} \text{FP}_{\varepsilon=0} T(1+u, 1+v; a) &= C_a - \gamma_1(a) \frac{d_1}{c_1} - \gamma_0(a) \frac{c_2}{c_1^2} \\ &\quad + \frac{1}{c_1 A} \left\{ \frac{c_2^2}{c_1^2} + \frac{B^2}{A^2} + \frac{c_2 B}{c_1 A} - \frac{c_3}{c_1} - \frac{C}{A} \right\}. \end{aligned} \quad (3.9)$$

Proof. The constant term of the holomorphic part is C_a . In $g_a(v)/u$, the constant terms are $-\gamma_0(a)c_2/c_1^2$ and $-\gamma_1(a)d_1/c_1$. Expanding both reciprocal factors in $1/[u(u+v)]$ to second order after factoring out $c_1 A \varepsilon^2$ gives the braces. Terms of order four in the curve cannot contribute. $\square$

For example, $u = \varepsilon$, $v = \varepsilon + \lambda\varepsilon^2$ gives

$$\text{FP } T(1 + \varepsilon, 1 + \varepsilon + \lambda\varepsilon^2; a) = C_a - \gamma_1(a) + \frac{\lambda^2}{8}. \quad (3.10)$$

Thus even curves with the same tangent as the diagonal have different finite parts. This is an explicit answer to the path-dependence question, not merely an assertion that multivariate regularization can be ambiguous. Even the same geometric diagonal, reparametrized by $u = v = \varepsilon + \lambda\varepsilon^2 + \mu\varepsilon^3$, has finite part

$$C_a - \gamma_1(a) - \lambda\gamma_0(a) + \frac{3\lambda^2 - 2\mu}{2}.$$

This also follows directly from the exact diagonal stuffle identity, and independently checks the cubic-jet terms in (3.9).

3.3 All regular Taylor coefficients in Stieltjes coordinates

The convergent sums $S_{m,n}(a)$ from (2.7) give a direct coordinate system for the complete holomorphic remainder.

Theorem 3.4 (The complete two-variable regular jet). *Write $H_a(u, v) = \sum_{m,n \geq 0} h_{m,n}(a) u^m v^n$. For all $m, n \geq 0$,*

$$\begin{aligned} h_{m,n}(a) &= \frac{(-1)^{m+n}}{m!n!} \left\{ \gamma_m(a)\gamma_n(a) + \frac{\gamma_{m+n+1}(a)}{n+1} - S_{m,n}(a) \right\} \\ &\quad - \frac{\zeta^{(m+n)}(2, a)}{2m!n!}. \end{aligned} \quad (3.11)$$

For a transverse straight direction, every regular coefficient is therefore explicit:

$$[\varepsilon^q]T(1 + c\varepsilon, 1 + d\varepsilon; a) = \frac{(-1)^{q+1}\gamma_{q+1}(a)d^{q+1}}{c(q+1)!} + \sum_{m+n=q} h_{m,n}(a)c^m d^n, \quad q \geq 0. \quad (3.12)$$

Proof. Normal convergence permits coefficient extraction in R_a . The coefficient of v^n in $E(v, x)$ is

$$\frac{(-1)^n}{n!} \left\{ \gamma_n(x) + \frac{\log^{n+1} x}{n+1} - \frac{\log^n x}{2x} \right\}.$$

Consequently $[u^m v^n]R_a = (-1)^{m+n} S_{m,n}/(m!n!)$. The coefficient of the divided difference in (3.6) is $-\binom{m+n+1}{n+1} g_{m+n+1}(a)$. Combining these two observations with the coefficients of $g_a(u)g_a(v)$ and $\zeta(2+u+v, a)/2$ proves (3.11). Substitution in (3.2) proves (3.12). $\square$

Taking symmetric coefficients in (3.3) and using (3.11) recovers (2.8). Thus the nonlinear Stieltjes summation law has two independent proofs: discrete telescoping and the meromorphic double-zeta stuffle identity. This compatibility is also an effective sign and normalization check.

4 Every directional Laurent coefficient of fully symmetrized depth

At depth greater than two, resolving one ordered multiple-zeta germ requires additional information. Complete symmetrization nevertheless admits an exact answer at every depth and every Laurent order. The algebraic input is Hoffman's classical symmetric-sum principle [8]; the present use records independent perturbation directions and the resulting Stieltjes coefficients explicitly.

For $a > 0$ and $d \geq 1$ set

$$Z_d(s_1, \dots, s_d; a) = \sum_{n_1 > \dots > n_d \geq 0} \prod_{i=1}^d (n_i + a)^{-s_i},$$

initially in its domain of absolute convergence. We sum over all *labelled* permutations,

$$\text{Sym } Z_d(\mathbf{s}; a) = \sum_{\sigma \in S_d} Z_d(s_{\sigma(1)}, \dots, s_{\sigma(d)}; a).$$

Repeated exponent values still occur with their permutation multiplicity. Let Π_d denote the set partitions of $\{1, \dots, d\}$.

Proposition 4.1 (Classical partition identity). *As an identity of meromorphic functions,*

$$\text{Sym } Z_d(\mathbf{s}; a) = \sum_{\pi \in \Pi_d} (-1)^{d-|\pi|} \prod_{B \in \pi} \left[(|B|-1)! \zeta \left(\sum_{i \in B} s_i, a \right) \right]. \quad (4.1)$$

Proof. First truncate all indices to $0 \leq n < N$. The symmetrized ordered sum is the sum over all labelled tuples with distinct indices. Inclusion–exclusion on equality patterns is Möbius inversion in the partition lattice. Its coefficient for a block B is $(-1)^{|B|-1}(|B|-1)!$. Each unrestricted equality block contributes $\sum_{n < N} (n+a)^{-\sum_{i \in B} s_i}$. This proves the finite identity. Letting $N \rightarrow \infty$ in the absolute-convergence region gives (4.1); meromorphic continuation gives the asserted identity. Alternatively, the right side itself supplies the continuation of the symmetrized function used below. $\square$

Theorem 4.2 (All-depth, all-order directional coefficient algorithm). *Let $c_1, \dots, c_d \in \mathbb{C} \setminus \{0\}$ and put $s_i = 1 + c_i \varepsilon$. For a block B , write $b = |B|$ and $c_B = \sum_{i \in B} c_i$. Define the coefficient array*

$$A_{\{i\}}(-1) = \frac{1}{c_i}, \quad A_{\{i\}}(k) = \frac{(-1)^k \gamma_k(a) c_i^k}{k!} \quad (k \geq 0), \quad (4.2)$$

$$A_B(k) = \frac{\zeta^{(k)}(b, a) c_B^k}{k!} \quad (k \geq 0, b \geq 2). \quad (4.3)$$

All other entries are zero. Then, for every integer $q \geq -d$,

$$\boxed{[\varepsilon^q] \text{Sym } Z_d(1 + c_1 \varepsilon, \dots, 1 + c_d \varepsilon; a) = \sum_{\pi \in \Pi_d} (-1)^{d-|\pi|} \prod_{B \in \pi} (|B| - 1)! \sum_{\substack{(k_B)_{B \in \pi} \\ \sum_B k_B = q}} \prod_{B \in \pi} A_B(k_B)}. \quad (4.4)$$

For each fixed d, q the displayed expression is finite. Vanishing block sums c_B with $|B| \geq 2$ cause no singularity. The constant coefficient uses only $\gamma_0(a), \dots, \gamma_{d-1}(a)$ and finitely many derivatives $\zeta^{(j)}(b, a)$ at integers $2 \leq b \leq d$.

Proof. For a singleton, the zeta factor in (4.1) has exactly the Laurent coefficients (4.2). For a block of size at least two it is regular at $\varepsilon = 0$, with Taylor coefficients (4.3). Multiplication proves (4.4). There are finitely many partitions, each index has lower bound -1 or 0 , and their sum is prescribed; hence only finitely many tuples can contribute. For $q = 0$, a singleton index is at most $d - 1$, which proves the Stieltjes-index assertion. $\square$

This is an explicit finite algorithm: enumerate set partitions, generate each block's coefficient list to the required degree, and multiply the short Laurent polynomials. No nested infinite sums or unspecified finite parts remain in the output.

4.1 Depth two and depth three in closed form

For depth two, the constant coefficient is

$$\text{FP Sym } Z_2 = \gamma_0(a)^2 - \zeta(2, a) - \gamma_1(a) \left(\frac{c_1}{c_2} + \frac{c_2}{c_1} \right), \quad (4.5)$$

in agreement with the sum of the two ordered values in (3.8) whenever both ordered restrictions are transverse. Formula (4.5) itself also applies when $c_1 + c_2 = 0$: the polar divisor of each ordered summand has canceled in the symmetrized meromorphic germ before restriction.

For depth three let $c_\Sigma = c_1 + c_2 + c_3$. Direct expansion gives

$$\begin{aligned} \text{FP Sym } Z_3 &= \gamma_0(a)^3 - \gamma_0(a)\gamma_1(a) \sum_{i \neq j} \frac{c_j}{c_i} + \frac{\gamma_2(a)}{2} \sum_{i=1}^3 \frac{c_i^2}{\prod_{j \neq i} c_j} \\ &\quad - 3\gamma_0(a)\zeta(2, a) - \zeta'(2, a) \sum_{i=1}^3 \frac{c_\Sigma - c_i}{c_i} + 2\zeta(3, a). \end{aligned} \quad (4.6)$$

At $c_1 = c_2 = c_3 = 1$ this becomes

$$\gamma_0(a)^3 - 6\gamma_0(a)\gamma_1(a) + \frac{3}{2}\gamma_2(a) - 3\gamma_0(a)\zeta(2, a) - 6\zeta'(2, a) + 2\zeta(3, a). \quad (4.7)$$

The corresponding single diagonal depth-three finite part is one sixth of (4.7); the factor $3!$ is required by our labelled convention.

4.2 Why coefficient extraction is not automatically a renormalization

The Laurent constant of a product is generally different from the product of the Laurent constants. For example,

$$\text{FP}_{\varepsilon=0}\{\zeta(1+c\varepsilon, a)\zeta(1+d\varepsilon, a)\} = \gamma_0(a)^2 - \gamma_1(a)(c/d + d/c), \quad (4.8)$$

whereas the product of the two individual constant coefficients is $\gamma_0(a)^2$. Therefore one cannot assert that a collection of directional finite parts preserves the stuffle algebra simply because the meromorphic functions satisfy stuffle identities. Algebra-preserving renormalizations require a compatible subtraction prescription; this is the subject of, for example, Guo and Zhang [10].

Theorems 3.1 and 4.2 settle two precise parts of the incoming questions: every ordered depth-two transverse direction, and every fully symmetrized direction at arbitrary depth. They do not give each individual ordered germ at depth three and higher, nor do they identify a canonical renormalization independent of a chosen path and subtraction convention.

5 All spectral derivatives at the moving Gamma zeros

The incoming *Nested Harmonic Jets* report [2] introduces a holomorphic spectral product, evaluates its first spectral derivative at each zero of its Gamma slice, and asks for an explicit construction at every derivative order. We give such a construction. Its coefficients have a finite triangular description by normalized Stieltjes antiderivatives. This solves the algorithmic part of that question; it gives a sufficient coordinate system rather than an arithmetic minimality theorem for constants.

The Weierstrass Gamma product, Bell polynomials, Stirling numbers, and Mellin transforms used below are classical. The contribution here is their specific combination into the all-order zero evaluation, its finite Stieltjes primitive closure, and a harmonic-number coefficient algorithm. The elementary digamma integral appearing at second order is not claimed as a new evaluation in isolation.

Fix $a > 0$, put $x_n = n + a$, and adopt

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{r \geq 0} \frac{(-1)^r \gamma_r(a)}{r!} u^r, \quad \gamma_r = \gamma_r(1), \quad \gamma_0(a) = -\psi(a).$$

Use the source product and its specified counterterm:

$$\mathcal{G}(s, z; a) = e^{z(\zeta(s, a) - 1/(s-1))} \prod_{n \geq 0} (1 + zx_n^{-s}) e^{-zx_n^{-s}}, \quad \Re s > \frac{1}{2}, \quad (5.1)$$

$$F_R(z; a) = \partial_s^R \mathcal{G}(s, z; a) \Big|_{s=1}. \quad (5.2)$$

The pole-subtracted expression in (5.1) is interpreted by its holomorphic extension at one. Local uniform convergence, together with the classical Gamma product, gives

$$\mathcal{G}(1, z; a) = \frac{\Gamma(a)}{\Gamma(a+z)}.$$

We evaluate F_R at the fixed argument $z = -a - m$. This is different from differentiating along the zero path $z = -(a+m)^s$, where the function is identically zero.

5.1 A finite primitive coordinate system

For $x \geq 0$, integers $r \geq 0$, and integers $k \geq 2$, define

$$L_r(x) = \sum_{n=1}^{\infty} \frac{\log^r(n+x) - \log^r n}{n}, \quad L_0(x) = 0, \quad (5.3)$$

$$M_{r,k}(x) = \sum_{n=1}^{\infty} \frac{\log^r(n+x)}{n^k}. \quad (5.4)$$

The convention for the zeroth power is one. The sums and their fixed derivatives converge locally uniformly: the summands of L_r , for $r > 0$, are $O(\log^{r-1} n/n^2)$ on a compact x -interval, and those of $M_{r,k}$ are $O(\log^r n/n^k)$. They also define holomorphic functions on $\mathbb{C} \setminus (-\infty, -1]$ using principal logarithms.

Put

$$T_r(x) = L_r(x) + \gamma_r - \gamma_r(1+x). \quad (5.5)$$

Theorem 5.1 (Finite Stieltjes primitive closure). *For $r \geq 1$ and $x > 0$,*

$$L'_r(x) = \frac{r}{x} T_{r-1}(x), \quad L_r(0) = 0, \quad (5.6)$$

$$M'_{r,k}(x) = r \left\{ \sum_{j=2}^k \frac{(-1)^{k-j}}{x^{k-j+1}} M_{r-1,j}(x) + \frac{(-1)^{k-1}}{x^k} T_{r-1}(x) \right\}, \quad (5.7)$$

$$M_{r,k}(0) = (-1)^r \zeta^{(r)}(k), \quad M_{0,k}(x) = \zeta(k). \quad (5.8)$$

The combined right sides have removable singularities at zero. Consequently, any finite rectangle of L_r and $M_{r,k}$ can be constructed by finitely many ordinary integrations from zero, starting with ordinary zeta derivatives and the generalized Stieltjes functions.

Proof. The defining limit for generalized Stieltjes constants gives

$$\begin{aligned} \sum_{n=1}^{\infty} \log^r(n+x) \left(\frac{1}{n} - \frac{1}{n+x} \right) &= L_r(x) + \gamma_r - \gamma_r(1+x) \\ &= T_r(x). \end{aligned} \quad (5.9)$$

To see the normalization explicitly, subtract the two cutoff sums for γ_r and $\gamma_r(1+x)$. Their integral counterterms differ by $[\log^{r+1}(N+1+x) - \log^{r+1}(N+1)]/(r+1)$, which tends to zero. The remaining difference is absolutely convergent. Differentiating (5.3) termwise and using $1/[n(n+x)] = x^{-1}(1/n - 1/(n+x))$ proves (5.6).

For $k \geq 2$, the elementary partial-fraction identity

$$\frac{1}{n^k(n+x)} = \sum_{j=2}^k \frac{(-1)^{k-j}}{x^{k-j+1} n^j} + \frac{(-1)^{k-1}}{x^k} \left(\frac{1}{n} - \frac{1}{n+x} \right) \quad (5.10)$$

and termwise differentiation of (5.4) prove (5.7). At zero, differentiation of the absolutely convergent Dirichlet series for $\zeta(k)$ gives (5.8). The original differentiated series are holomorphic near zero. Thus the combined rational expressions on the right of (5.6)–(5.7) have removable singularities there. Integrating those combined expressions proves the last assertion by induction on r . $\square$

One must integrate the combined right side of (5.7). Its individual displayed terms can have nonintegrable poles at zero, although their sum is regular.

In particular, the first coordinate is the ordinary digamma primitive

$$D(x) := L_1(x) = \sum_{n \geq 1} \frac{\log(1 + x/n)}{n} = \int_0^x \frac{\psi(1+t) + \gamma}{t} dt. \quad (5.11)$$

The integrand takes the limiting value $\zeta(2)$ at zero. Higher L_r are obtained from

$$L_r(x) = r \int_0^x \frac{L_{r-1}(t) + \gamma_{r-1} - \gamma_{r-1}(1+t)}{t} dt.$$

For example

$$L'_2(x) = \frac{2}{x} (D(x) + \gamma_1 - \gamma_1(1+x)), \quad (5.12)$$

$$M'_{2,2}(x) = \frac{2}{x} M_{1,2}(x) - \frac{2}{x^2} (D(x) + \gamma_1 - \gamma_1(1+x)). \quad (5.13)$$

5.2 The zero-factor theorem

Let $\{^r_k\}$ be a Stirling number of the second kind and set

$$A_{r,k} = (k-1)! \left\{^r_k\right\}. \quad (5.14)$$

Let B_j be the Bernoulli numbers with $B_1 = -1/2$. Define the complete exponential Bell polynomials by

$$\exp \left(\sum_{j \geq 1} c_j \frac{u^j}{j!} \right) = \sum_{n \geq 0} Y_n(c_1, \dots, c_n) \frac{u^n}{n!}.$$

Theorem 5.2 (All fixed-argument spectral jets at a Gamma zero). *Let $a > 0$, $m \in \mathbb{Z}_{\geq 0}$, and set*

$$x = a + m, \quad \ell = \log x, \quad h_0 = x\Gamma(a)(-1)^m m!.$$

For $r \geq 1$ define the finite expression

$$b_r = (-1)^{r+1} \left\{ x(\gamma_r + L_r(x)) + \sum_{k=2}^r A_{r,k} x^k M_{r,k}(x) + \sum_{j=1}^m \log^r(x-j) \sum_{k=1}^r A_{r,k} \left(-\frac{x}{j}\right)^k \right\}, \quad (5.15)$$

$$c_r = b_r + \frac{B_r}{r} \ell^r. \quad (5.16)$$

Empty sums vanish. Then, for every $R \geq 1$,

$$\boxed{F_R(-x; a) = R h_0 \ell Y_{R-1}(c_1, \dots, c_{R-1})}. \quad (5.17)$$

At $x = 1$, every $F_R(-1; 1)$ is zero. For $x \neq 1$, the formula requires only the $R - 1$ logarithmic jets $b_1, \dots, b_{R-1}$.

Proof. Remove the unique vanishing factor:

$$\mathcal{G}(1+u, -x; a) = (1 - e^{-\ell u})\mathcal{H}(1+u),$$

where $\mathcal{H}$ is holomorphic and nonzero near one. At $s = 1$, the Gamma slice has derivative $\Gamma(a)(-1)^m m!$ in z at $-x$. The removed factor is $(z+x)/x$ on that slice, so $\mathcal{H}(1) = h_0$. Equivalently, for a cutoff $N > m$, the zero-removed value is

$$\mathcal{H}_N(1) = h_0(N+a)^x \frac{\Gamma(N-m)}{\Gamma(N+a)}, \quad (5.18)$$

which tends to h_0 by the Gamma ratio asymptotic.

The source cutoff is

$$\mathcal{G}_N(s, -x; a) = e^{xC_N(s;a)} \prod_{n < N} (1 - xx_n^{-s}), \quad C_N(s; a) = \frac{1 - (N+a)^{1-s}}{s-1}.$$

Its derivatives satisfy $\partial_s^r C_N(1; a) = (-1)^r \log^{r+1}(N+a)/(r+1)$. For integer $r \geq 1$,

$$\partial_s^r \log(1 - xx_n^{-s})|_{s=1} = (-1)^{r+1} \log^r x_n \operatorname{Li}_{1-r}(x/x_n). \quad (5.19)$$

Only integer nonpositive polylogarithm orders occur here, hence these are rational functions; the formula is valid even for $n < m$, when the argument exceeds one. The rational identity

$$\operatorname{Li}_{1-r}(q) = \sum_{k=1}^r A_{r,k} \left(\frac{q}{1-q} \right)^k \quad (5.20)$$

follows either by applying $q\partial_q$ repeatedly to $q/(1-q)$ or from the Stirling recurrence. Thus

$$\partial_s^r \log \mathcal{H}_N(s)|_{s=1} = (-1)^r \left\{ \frac{x}{r+1} \log^{r+1}(N+a) - \sum_{\substack{0 \leq n < N \\ n \neq m}} \log^r(n+a) \sum_{k=1}^r \frac{A_{r,k} x^k}{(n-m)^k} \right\}. \quad (5.21)$$

The $n < m$ terms become the finite j -sum in (5.15). Set $n = m + j$ in the remaining terms. Since $A_{r,1} = 1$,

$$\lim_{J \rightarrow \infty} \left\{ \sum_{j=1}^J \frac{\log^r(j+x)}{j} - \frac{\log^{r+1}(J+1+x)}{r+1} \right\} = \gamma_r + L_r(x).$$

Indeed, subtract the analogous limit for γ_r ; the change in its logarithmic counterterm tends to zero. The sums for $k \geq 2$ converge to $M_{r,k}(x)$. Local uniform convergence of the zero-removed products justifies passage to their logarithmic derivatives, proving $b_r = \partial_s^r \log \mathcal{H}(1)$ and (5.15).

For $\ell \neq 0$, the Bernoulli generating function yields, near $u = 0$,

$$\log \frac{1 - e^{-\ell u}}{\ell u} = \sum_{r \geq 1} \frac{B_r \ell^r}{r} \frac{u^r}{r!}. \quad (5.22)$$

Consequently

$$\mathcal{G}(1+u, -x; a) = h_0 \ell u \exp \left(\sum_{r \geq 1} c_r \frac{u^r}{r!} \right).$$

Bell coefficient extraction proves (5.17). If $\ell = 0$, the removed factor $1 - e^{-\ell u}$ vanishes identically, proving the stated exceptional case without division by zero. $\square$

The first three nontrivial rows are

$$F_1(-x; a) = h_0 \ell, \quad (5.23)$$

$$F_2(-x; a) = h_0(2\ell b_1 - \ell^2), \quad (5.24)$$

$$F_3(-x; a) = h_0\{3\ell(b_1^2 + b_2) - 3\ell^2 b_1 + \ell^3\}. \quad (5.25)$$

The first is the source's already proved zero-flow formula. The second has the explicit Gamma–Stieltjes–digamma–primitive evaluation

$$F_2(-x; a) = h_0 \left[2x\ell \left\{ \gamma_1 + D(x) - \sum_{j=1}^m \frac{\log(x-j)}{j} \right\} - \ell^2 \right]. \quad (5.26)$$

For the next row,

$$b_2 = - \left[x(\gamma_2 + L_2(x)) + x^2 M_{2,2}(x) + \sum_{j=1}^m \log^2(x-j) \left(-\frac{x}{j} + \frac{x^2}{j^2} \right) \right]. \quad (5.27)$$

For fixed $R \geq 2$, Theorem 5.1 supplies a finite construction of all entries needed in (5.17). It uses ordinary Stieltjes constants only through γ_{R-1} , Stieltjes *functions* $\gamma_j(1+t)$ only through $j = R-2$, and finitely many zeta derivatives $\zeta^{(r)}(k)$ with $0 \leq r \leq R-1$ and $2 \leq k \leq R-1$. For $R \geq 2$, the highest ordinary Stieltjes constant occurs linearly, with coefficient

$$[\gamma_{R-1}]F_R(-x; a) = (-1)^R R x h_0 \log x \quad (5.28)$$

in this explicit coordinate formula. This is a statement about the displayed construction. It does not assert arithmetic independence of Stieltjes constants from zeta derivatives or their integrals.

5.3 Polylogarithmic Mellin realization

The coordinate system also has a direct polylogarithm interpretation. Set

$$J_k(v, x) = \sum_{n=1}^{\infty} \frac{1}{n^k (n+x)^v}. \quad (5.29)$$

For $x \geq 0$ and $\Re v > 0$, termwise integration of the absolutely convergent polylogarithm series gives

$$J_k(v, x) = \frac{1}{\Gamma(v)} \int_0^{\infty} t^{v-1} e^{-xt} \text{Li}_k(e^{-t}) dt, \quad k \geq 1. \quad (5.30)$$

For $k \geq 2$ its Dirichlet series is holomorphic near $v = 0$, and

$$M_{r,k}(x) = (-1)^r \partial_v^r J_k(0, x).$$

For $k = 1$, the difference

$$J_1(v, x) - \zeta(1+v) = \sum_{n \geq 1} \frac{(n+x)^{-v} - n^{-v}}{n}$$

is holomorphic near zero because its summands are $O(n^{-2-\Re v})$ there, uniformly on compact sets. It follows that

$$J_1(v, x) = \frac{1}{v} + \sum_{r \geq 0} \frac{(-1)^r (\gamma_r + L_r(x))}{r!} v^r. \quad (5.31)$$

Thus the exact Gamma-zero derivatives are finite combinations of resonant Laurent coefficients and regular spectral derivatives of the elementary polylogarithmic Mellin transform (5.30). This is a specific shifted Dirichlet series, not an identification of its Laurent coefficients with the ordinary Hurwitz Stieltjes constants.

5.4 Generalized harmonic numbers give every coefficient

Let $e_j(h)$ be defined by

$$\prod_{n=1}^h (1 + u/n) = \sum_{j=0}^h e_j(h) u^j, \quad e_0(h) = 1. \quad (5.32)$$

These are polynomials in the generalized harmonic numbers $H_h^{(j)} = \sum_{n=1}^h n^{-j}$:

$$e_1(h) = H_h, \quad e_2(h) = \frac{1}{2}(H_h^2 - H_h^{(2)}), \quad \sum_j e_j(h) u^j = \exp \left(\sum_{j \geq 1} \frac{(-1)^{j-1} H_h^{(j)}}{j} u^j \right).$$

For $r \geq 1$, $h \geq 1$, and $k \geq 1$, put

$$Q_{r,k,h}(q) = \sum_{j=1}^{\min(r,h)} \frac{r!}{(r-j)!} e_{j-1}(h-1) \zeta^{(r-j)}(k+h, q). \quad (5.33)$$

Theorem 5.3 (Harmonic–Hurwitz coefficient formulas). *Let $r \geq 1$, let $N \geq 0$ be an integer, and let $x \in \mathbb{C} \setminus (-\infty, -1]$ satisfy $|x| < N+1$. Then*

$$L_r(x) = \sum_{n=1}^N \frac{\log^r(n+x) - \log^r n}{n} + (-1)^r \sum_{h \geq 1} \frac{(-x)^h}{h} Q_{r,1,h}(N+1), \quad (5.34)$$

$$\begin{aligned} M_{r,k}(x) &= \sum_{n=1}^N \frac{\log^r(n+x)}{n^k} + (-1)^r \zeta^{(r)}(k, N+1) \\ &\quad + (-1)^r \sum_{h \geq 1} \frac{(-x)^h}{h} Q_{r,k,h}(N+1), \quad k \geq 2. \end{aligned} \quad (5.35)$$

The series converge normally on compact subsets of $|x| < N+1$. Every positive x is covered after a finite cutoff. The only harmonic orders needed for a spectral derivative of order r are at most $r-1$.

Proof. Split (5.29) at N and use the binomial expansion in the tail:

$$\sum_{n > N} \frac{1}{n^k (n+x)^v} = \sum_{h \geq 0} \frac{(-1)^h (v)_h x^h}{h!} \zeta(k+v+h, N+1).$$

For $|x| < N+1$, the double series and its fixed v derivatives converge normally on the appropriate local domain; the $h=0, k=1$ pole is treated separately. Since

$$\frac{(v)_h}{h!} = \frac{v}{h} \prod_{j=1}^{h-1} (1 + v/j) = \frac{1}{h} \sum_{j=1}^h e_{j-1}(h-1) v^j,$$

Leibniz differentiation at zero gives the coefficient $Q_{r,k,h}(N+1)/h$. Formula (5.31) or the derivative description of $M_{r,k}$ then proves (5.34)–(5.35). The finite harmonic expression follows from (5.32). $\square$

For example, for $|x| < 1$,

$$D(x) = \sum_{h \geq 1} \frac{(-1)^{h+1} x^h}{h} \zeta(h+1), \quad (5.36)$$

$$L_2(x) = 2 \sum_{h \geq 1} \frac{(-x)^h}{h} \{\zeta'(h+1) + H_{h-1} \zeta(h+1)\}, \quad (5.37)$$

$$M_{2,2}(x) = \zeta''(2) + 2 \sum_{h \geq 1} \frac{(-x)^h}{h} \{\zeta'(h+2) + H_{h-1} \zeta(h+2)\}. \quad (5.38)$$

Precise remaining questions. The finite coordinate construction is complete, but whether some of its nontrivial fixed-argument integrals reduce to a smaller field of special values remains open. The first test is $D(x)$ at a positive integer or a nonintegral rational point. No failure of a symbolic or numerical search would prove non-reducibility. A second target is a global description of the individual higher spectral jets as entire functions of z , with these Gamma-zero evaluations as interpolation data. A third target is the compatible construction for several simultaneously moving zero factors in multicolour spectral products.

6 A two-shift Lerch reflection for inverse-color doubles

The manuscript already proves an all-weight reduction of $\text{Li}_{a,b}(z, z^{-1})$ to one-direction double polylogarithms. Its explicit mixed parity formula treats $b = 1$; the general parity component also follows by combining that reduction with the all-index one-direction parity theorem. Thus an extension from $b = 1$ alone must not be advertised as a new reducibility principle. The established multiple-polylogarithm parity theory includes Panzer’s general theorem and explicit depth-two formulas [11]. More recently, Xu’s 2026 preprint *Symmetry Results for Cyclotomic Multiple Hurwitz Zeta Values via Contour Integrals*, arXiv:2602.10391 [13], proves broad shifted cyclotomic symmetry identities. Its index order is increasing, whereas the manuscript uses decreasing indices.

The purpose of the following calculation is more specific: it gives a compact two-shift identity that is an ordinary holomorphic identity in the full disk of a continuous color variable. Its arbitrary-index version follows from a single first-order seed by parameter differentiation. It also gives complete reductions, rather than only paired reductions, on the complementary-shift diagonal. These are useful explicit additions to the manuscript. No claim of worldwide priority is made for the underlying symmetry.

6.1 Definitions and analytic domain

For integers $a, b \geq 1$, define

$$\begin{aligned} \mathcal{M}_{a,b}(z; u, v) &= \sum_{n > m \geq 0} \frac{z^{n-m}}{(n+u)^a (m+v)^b} = \sum_{h \geq 1} z^h c_{a,b}(h; u, v), \\ c_{a,b}(h; u, v) &= \sum_{m \geq 0} \frac{1}{(m+h+u)^a (m+v)^b}. \end{aligned} \quad (6.1)$$

We initially work on

$$|z| < 1, \quad 0 < \operatorname{Re} u < 1, \quad 0 < \operatorname{Re} v < 1. \quad (6.2)$$

All powers are integer powers, so there is no denominator branch choice. The Lerch function is

$$\Phi(z, s, q) = \sum_{m \geq 0} \frac{z^m}{(m+q)^s} \quad (|z| < 1, \operatorname{Re} q > 0).$$

Put

$$C_1(q) = \pi \cot \pi q, \quad C_r(q) = \zeta(r, q) + (-1)^r \zeta(r, 1-q) \quad (r \geq 2). \quad (6.3)$$

Equivalently,

$$C_r(q) = \frac{(-1)^{r-1}}{(r-1)!} \frac{d^{r-1}}{dq^{r-1}} (\pi \cot \pi q). \quad (6.4)$$

Thus every C_r is elementary in trigonometric functions, and can equally be expressed through polygamma functions.

The series in (6.1) converges absolutely and normally on compact subsets of (6.2). Indeed, uniformly on compact shift sets,

$$|c_{a,b}(h; u, v)| \leq C \sum_{m \geq 0} \frac{1}{(m+h+1)^a (m+1)^b} = O\left(\frac{1 + \log(h+1)}{(h+1)^a}\right).$$

The same estimates hold after any fixed number of shift derivatives. In particular all parameter differentiations below are legitimate holomorphic differentiations, not formal manipulations of a divergent series.

6.2 The first-order seed

Theorem 6.1 (Two-shift seed). *On (6.2),*

$$\begin{aligned} & \mathcal{M}_{1,1}(z; u, v) + \mathcal{M}_{1,1}(z; 1-v, 1-u) \\ &= z\Phi(z, 1, u)\Phi(z, 1, 1-v) + (C_1(v) - C_1(u))z\Phi(z, 1, 1+u-v). \end{aligned} \quad (6.5)$$

All apparent confluences at $u = v$ are removable; in fact no divided difference is needed in the displayed formula.

Proof. Fix a positive integer h . The absolutely convergent bilateral sum

$$B_{1,1}(h; u, v) = \sum_{m \in \mathbb{Z}} \frac{1}{(m+h+u)(m+v)}$$

has three disjoint ranges. The range $m \geq 0$ is $c_{1,1}(h; u, v)$. In the range $m \leq -h-1$, set $m = -h-1-\ell$; its contribution is $c_{1,1}(h; 1-v, 1-u)$. In the middle range $-h \leq m \leq -1$, write $m = -\ell-1$ and $j = h-1-\ell$. Its contribution is

$$- \sum_{j+\ell=h-1} \frac{1}{(j+u)(\ell+1-v)}.$$

On the other hand, with $d = h+u-v$,

$$\frac{1}{(m+h+u)(m+v)} = \frac{1}{d} \left(\frac{1}{m+v} - \frac{1}{m+h+u} \right).$$

Symmetric summation gives $B_{1,1}(h; u, v) = (C_1(v) - C_1(u))/(h+u-v)$. The shift by the integer h does not change a symmetric cotangent sum: the finitely shifted endpoint terms tend to zero. Multiply by z^h and sum over $h \geq 1$. The middle range is the coefficient convolution of the two Lerch series, and the bilateral term sums to the second term in (6.5). All resulting h -series converge absolutely for $|z| < 1$. $\square$

6.3 Every pair of indices, in two equivalent forms

Theorem 6.2 (Arbitrary-index reflection). *Let $a, b \geq 1$ and $w = a + b$. Set $q = 1 + u - v$. Then*

$$\begin{aligned} & \mathcal{M}_{a,b}(z; u, v) + (-1)^w \mathcal{M}_{b,a}(z; 1 - v, 1 - u) \\ &= z \left[(-1)^b \sum_{r=1}^a \binom{w-r-1}{b-1} C_r(u) \Phi(z, w-r, q) \right. \\ & \quad + \sum_{r=1}^b (-1)^{b-r} \binom{w-r-1}{a-1} C_r(v) \Phi(z, w-r, q) \\ & \quad \left. - (-1)^b \Phi(z, a, u) \Phi(z, b, 1 - v) \right]. \end{aligned} \quad (6.6)$$

Equivalently, the right side is

$$\frac{(-1)^w}{(a-1)!(b-1)!} \partial_u^{a-1} \partial_v^{b-1} [z \Phi(z, 1, u) \Phi(z, 1, 1 - v) + (C_1(v) - C_1(u)) z \Phi(z, 1, 1 + u - v)]. \quad (6.7)$$

Proof. For the derivative form, apply the stated mixed derivative to (6.5). On the first summand on its left, the derivative contributes the sign $(-1)^{a+b-2} = (-1)^w$ and the factorial factor. On the reflected summand both chain-rule signs cancel the signs from differentiating a reciprocal, and the two indices are exchanged. This gives (6.7) with the precise sign shown.

For an independent coefficient proof of (6.6), put $d = h + u - v$. The rational partial-fraction identity is

$$\begin{aligned} \frac{1}{(m+h+u)^a(m+v)^b} &= (-1)^b \sum_{r=1}^a \binom{w-r-1}{b-1} \frac{1}{d^{w-r}(m+h+u)^r} \\ & \quad + \sum_{r=1}^b (-1)^{b-r} \binom{w-r-1}{a-1} \frac{1}{d^{w-r}(m+v)^r}. \end{aligned} \quad (6.8)$$

It follows either by expanding at its two poles or by subtracting the two sides, observing that the difference has no finite poles and vanishes at infinity. Since $\text{Re } d > 0$, no collision occurs in the domain under consideration.

Sum over $m \in \mathbb{Z}$. For $r \geq 2$, the sum of each reciprocal power is $C_r(u)$ or $C_r(v)$. The $r = 1$ terms are evaluated by a common symmetric cutoff. Their principal parts at infinity cancel, because

$$\binom{w-2}{a-1} = \binom{w-2}{b-1}.$$

The original bilateral sum is absolutely convergent, so this use of a common cutoff changes no value. Split it into the same three ranges as in the first-order proof. The negative range is now $(-1)^w c_{b,a}(h; 1 - v, 1 - u)$, and the middle range is

$$(-1)^b \sum_{j+\ell=h-1} (j+u)^{-a} (\ell+1-v)^{-b}.$$

Multiplication by z^h and summation gives (6.6). □

6.4 Shift derivatives and normalized primitives

The whole family is closed under arbitrary shift derivatives:

$$\partial_u^r \partial_v^s \mathcal{M}_{a,b}(z; u, v) = (-1)^{r+s} (a)_r (b)_s \mathcal{M}_{a+r, b+s}(z; u, v). \quad (6.9)$$

Accordingly, for $a \geq 2$ a primitive in u is $-\mathcal{M}_{a-1,b}/(a-1)$, and for $b \geq 2$ a primitive in v is $-\mathcal{M}_{a,b-1}/(b-1)$. At the index-one boundary one should normalize the primitive before summing. For a path between u_0 and u_1 inside the right half-plane,

$$\int_{u_0}^{u_1} \mathcal{M}_{1,b}(z; u, v) du = \sum_{n>m \geq 0} \frac{z^{n-m}}{(m+v)^b} \log \frac{n+u_1}{n+u_0}. \quad (6.10)$$

Here the logarithm is the difference of the common analytic logarithms along the path. For $|z| < 1$ the normalized series is absolutely convergent, including $b = 1$, because the logarithmic difference is $O(n^{-1})$ on compact endpoint sets. The unnormalized series with $\log(n+u)$ alone need not converge when $b = 1$.

There is also a precise spectral-derivative interpretation. Define the termwise difference

$$\Delta \mathcal{M}_{s,b} = \sum_{n>m \geq 0} \frac{z^{n-m}}{(m+v)^b} ((n+u_1)^{-s} - (n+u_0)^{-s}).$$

It is holomorphic near $s = 0$, even when the two separate sums there are divergent, and the right side of (6.10) is $-\partial_s \Delta \mathcal{M}_{s,b}|_{s=0}$. This statement uses subtraction before continuation. It does not assert that the integer-index finite sum (6.6) can itself be differentiated in a noninteger exponent.

Boundary values. The identity extends to each $|z| = 1$, $z \neq 1$, as an ordinary convergent identity. To make this statement independent of an unspecified Abel prescription, note that on compact shift sets $c_{a,b}(h; u, v) \rightarrow 0$ and

$$c_{a,b}(h+1; u, v) - c_{a,b}(h; u, v) = O\left(\frac{1 + \log(h+1)}{(h+1)^{a+1}}\right).$$

The sum of these differences converges absolutely. Summation by parts therefore proves convergence uniformly on closed arcs that avoid 1. The same argument applies to every Lerch term on the right, whose parameter has positive real part, and to fixed shift derivatives. Taking a radial limit of the disk identity proves the boundary identity in the ordinary gap-ordered series convention.

It also agrees with the original outer-index order. Let O_N be the sum in (6.1) restricted to $n \leq N$, and let $D_N = \sum_{h=1}^N z^h c_{a,b}(h; u, v)$. Partial summation of the finite h -sum gives, uniformly for shifts in a compact subset of the strip,

$$|D_N - O_N| \leq \frac{C}{|1-z|} \left(N^{-a} (1 + \log(N+1)) + \sum_{m>N} m^{-a-b} \right) \rightarrow 0.$$

For $m \leq N$, the omitted h -range starts past $N-m$, so its denominator $(m+h+u)^a$ is of size at least N^a ; for $m > N$, the full finite h -sum is bounded by a constant times m^{-a} . These are exactly the two terms in the displayed bound. Thus no conditional rearrangement is hidden in the boundary statement. If $a, b \geq 2$, all terms are also absolutely convergent at $z = 1$, so that endpoint may then be included.

6.5 Complete reductions at complementary shifts

The reflection becomes a full evaluation when the two exponents agree and $v = 1 - u$.

Corollary 6.3 (Complementary diagonal). *For $a \geq 1$, $|z| < 1$ and $0 < \operatorname{Re} u < 1$,*

$$\mathcal{M}_{a,a}(z; u, 1 - u) = (-1)^a z \left[\sum_{r=1}^a \binom{2a-r-1}{a-1} C_r(u) \Phi(z, 2a-r, 2u) - \frac{1}{2} \Phi(z, a, u)^2 \right]. \quad (6.11)$$

In particular,

$$\mathcal{M}_{1,1}(z; u, 1 - u) = \frac{z}{2} \Phi(z, 1, u)^2 - \pi \cot(\pi u) z \Phi(z, 1, 2u), \quad (6.12)$$

and

$$\mathcal{M}_{2,2}(z; u, 1 - u) = z \left[2\pi \cot(\pi u) \Phi(z, 3, 2u) + \pi^2 \csc^2(\pi u) \Phi(z, 2, 2u) - \frac{1}{2} \Phi(z, 2, u)^2 \right]. \quad (6.13)$$

Proof. The reflected double on the left of (6.6) equals the original one. On the right, use $C_r(1-u) = (-1)^r C_r(u)$ and divide by two. $\square$

The $a = 1$ identity also exposes the generalized-harmonic content. With

$$H_h(u) = \psi(h+u) - \psi(u) = \sum_{j=0}^{h-1} \frac{1}{j+u},$$

partial fractions give

$$c_{1,1}(h; u, 1 - u) = \frac{H_h(u) - \pi \cot \pi u}{h + 2u - 1}.$$

Consequently (6.12) is equivalent to

$$\sum_{h \geq 1} \frac{z^h H_h(u)}{h + 2u - 1} = \frac{z}{2} \Phi(z, 1, u)^2. \quad (6.14)$$

This last elementary convolution identity can be proved directly by partial fractions in the coefficient of z^{h-1} in the square. That observation clarifies the strength of the claim: the result is an exact complete reduction, but the first-order instance already has an elementary coefficient explanation.

6.6 The half-shift diagonal in classical polylogarithms

Put

$$\chi_s(x) = \frac{\operatorname{Li}_s(x) - \operatorname{Li}_s(-x)}{2}.$$

For $z = x^2$ with $|x| < 1$,

$$\Phi(z, s, \tfrac{1}{2}) = \frac{2^s}{x} \chi_s(x), \quad C_{2r}(\tfrac{1}{2}) = 2(2^{2r} - 1)\zeta(2r), \quad C_{2r+1}(\tfrac{1}{2}) = 0.$$

The dependence on the chosen square root cancels in all products. Equation (6.11) gives the all-order identity

$$\mathcal{M}_{a,a}(z; \tfrac{1}{2}, \tfrac{1}{2}) = (-1)^a \left[2 \sum_{r=1}^{\lfloor a/2 \rfloor} (2^{2r} - 1) \binom{2a - 2r - 1}{a - 1} \zeta(2r) \text{Li}_{2a-2r}(z) - 2^{2a-1} \chi_a(\sqrt{z})^2 \right]. \quad (6.15)$$

The first four cases are

$$\mathcal{M}_{1,1}(z; \tfrac{1}{2}, \tfrac{1}{2}) = 2 \operatorname{artanh}^2 \sqrt{z}, \quad (6.16)$$

$$\mathcal{M}_{2,2}(z; \tfrac{1}{2}, \tfrac{1}{2}) = \pi^2 \text{Li}_2(z) - 8 \chi_2(\sqrt{z})^2, \quad (6.17)$$

$$\mathcal{M}_{3,3}(z; \tfrac{1}{2}, \tfrac{1}{2}) = 32 \chi_3(\sqrt{z})^2 - 3\pi^2 \text{Li}_4(z), \quad (6.18)$$

$$\mathcal{M}_{4,4}(z; \tfrac{1}{2}, \tfrac{1}{2}) = 10\pi^2 \text{Li}_6(z) + \frac{\pi^4}{3} \text{Li}_4(z) - 128 \chi_4(\sqrt{z})^2. \quad (6.19)$$

Here each left side is a two-index series at a variable argument; the displayed equality is a reduction of that function. At individual special points there may be additional elementary stuffle explanations, which should be recorded rather than presented as new arithmetic phenomena.

6.7 A classical antiderivative and an exact alternating moment

At the half shift,

$$H_h(\tfrac{1}{2}) = 2H_{2h} - H_h.$$

Hence (6.16) says

$$\sum_{h \geq 1} \frac{(2H_{2h} - H_h)z^h}{h} = 2 \operatorname{artanh}^2 \sqrt{z}. \quad (6.20)$$

Integrating once against dz/z is a useful explicit antiderivative. For $0 < z < 1$ put $x = (1 - \sqrt{z})/(1 + \sqrt{z})$. Then

$$\begin{aligned} \sum_{h \geq 1} \frac{(2H_{2h} - H_h)z^h}{h^2} &= \frac{7}{2} \zeta(3) - \log^2 x \log \frac{1+x}{1-x} \\ &\quad + 4 \log x \chi_2(x) - 4 \chi_3(x). \end{aligned} \quad (6.21)$$

To prove it, write the left side as $4 \int_0^{\sqrt{z}} \operatorname{artanh}^2 t \, dt/t$ and substitute $x = (1 - t)/(1 + t)$. An antiderivative of $\log^2 x ((1 - x)^{-1} + (1 + x)^{-1})$ is

$$\log^2 x \log \frac{1+x}{1-x} - 4 \log x \chi_2(x) + 4 \chi_3(x).$$

At $x = 1$ its value is $4\chi_3(1) = 7\zeta(3)/2$, fixing the constant. The real interval formulation avoids any hidden logarithm branches; continuation elsewhere is along the corresponding common path.

At the alternating endpoint one obtains

$$\sum_{h \geq 1} \frac{(-1)^h (2H_{2h} - H_h)}{h^2} = \frac{7}{2} \zeta(3) - 2\pi G, \quad G = \beta(2). \quad (6.22)$$

An entirely real proof follows from $-4 \int_0^1 \arctan^2 t \, dt/t$; alternatively continue (6.21) to $z = -1$, where $x = -i$ and the imaginary π^3 terms cancel. The series is absolutely convergent. This is a checkable example in the existing Euler-sum vocabulary, not a claim that this classical type of special value is new.

6.8 Rational shifts and finite root-of-unity filters

For integers $1 \leq p \leq q$, put $\omega = e^{2\pi i/q}$ and choose x with $x^q = z$. The residue-class filter gives

$$\Phi(z, s, p/q) = q^{s-1} x^{-p} \sum_{j=0}^{q-1} \omega^{-pj} \text{Li}_s(\omega^j x). \quad (6.23)$$

For integer $s \geq 1$ this follows directly by expanding the polylogarithms and projecting onto exponents congruent to p modulo q . For larger positive rational shifts, first use

$$\Phi(z, s, q_0 + N) = z^{-N} \left(\Phi(z, s, q_0) - \sum_{j=0}^{N-1} \frac{z^j}{(j + q_0)^s} \right).$$

All apparent singularities at $z = 0$ are removable. Thus rational instances of (6.6) and (6.11) reduce to finite expressions involving ordinary polylogarithms at algebraically related arguments and elementary cotangent derivatives. No extra depth is introduced by evaluating the single Lerch functions.

6.9 An independent integral check and reproducibility

For $a, b \geq 1$, $\text{Re } u, \text{Re } v > 0$ and $|z| < 1$,

$$\mathcal{M}_{a,b}(z; u, v) = \frac{z}{(a-1)!} \int_0^1 \frac{t^u (-\log t)^{a-1} \Phi(t, b, v)}{1-zt} dt. \quad (6.24)$$

Insert the Mellin integral for $(n+u)^{-a}$ and sum the geometric series in $h = n - m$. This proves the formula by absolute convergence. At a fixed $|z| = 1$, $z \neq 1$, its denominator stays away from zero on $[0, 1]$; the possible logarithmic singularity of $\Phi(t, 1, v)$ at $t = 1$ is integrable. The boundary formula follows by dominated convergence and the preceding summation-by-parts argument.

The supplied script `check_shifted_gap.py` evaluates this integral after $t = e^{-y}$, using finite root-of-unity filters for rational inner shifts. It compares that independent quadrature with (6.6) for unequal shifts and complex colors, and checks the first three half-diagonal formulas. At 60 decimal working digits, the largest observed absolute residual among the four general reflection checks is 8.33×10^{-60} . These checks are numerical diagnostics, not rigorous error enclosures; the proofs are the bilateral splitting and partial-fraction arguments above.

6.10 Precise continuation questions

1. Compare (6.6) term by term with the depth-two specialization of Xu's shifted cyclotomic symmetry formulas, recording the increasing/decreasing index conversion and color normalization. The existence of a broad earlier theorem should remain explicit.
2. Determine which non-diagonal parameter specializations make the reflected double equal to, or otherwise independently reducible from, the original. Complementary shifts with equal exponents give the complete reduction proved here.
3. Develop closed primitives of the complementary diagonal for $a \geq 2$ in a specified classical or multiple-polylogarithm basis. The half-shift first-order primitive (6.21) supplies a nontrivial normalization check.

4. Study noninteger spectral indices without differentiating the integer-index partial-fraction formula as though its summation range were analytic. Such an extension needs a separate contour or Mellin argument; the integer formula alone does not justify it.
5. Continue in the shifts beyond the safe strip while tracking the finitely many terms crossed when a shifted lattice point passes zero. This should be a meromorphic identity with explicit finite corrections, not an unqualified replacement of ordinary sums by principal values.

7 Local identities for every derivative of a spectral product

There is a useful second application of the distinction between a multivariate germ and its one-dimensional restrictions. For polynomially weighted products of shifted linear factors, every fixed spectral derivative is a polynomial in the *normalized factor exponents*. Its highest-degree component is local and explicitly computable. This produces finite identities between higher zeta derivatives even when the individual values do not have short closed forms.

The first-derivative multiplicative anomaly is classical in substance; Dowker treats linear factors and polynomial multiplicities [14], and Gürel gives general regularized-product formulas [15], extending the Lerch–Mizuno setting [16]. We retain that attribution. The all-order germ, the normalized-weight polynomial formulation, and its explicit finite-difference consequences are the proved extension developed here; the literature review does not establish a priority claim for them.

7.1 An exact multivariate meromorphic germ

Let P be a nonzero polynomial of degree D , let $a_1, \dots, a_m > 0$, and initially define

$$\mathcal{Z}_P(\mathbf{u}; \mathbf{a}) = \sum_{n \geq 0} P(n) \prod_{j=1}^m (n + a_j)^{-u_j}, \quad U = \sum_{j=1}^m u_j. \quad (7.1)$$

Absolute convergence requires $\operatorname{Re} U > D + 1$; no separate positivity of the u_j is needed. Choose $b > 0$ such that $\max_j |a_j - b| < b$, for example $b = \max_j a_j$. Write

$$d_j = a_j - b, \quad p_k = \frac{P^{(k)}(-b)}{k!}, \quad P(n) = \sum_{k=0}^D p_k (n + b)^k, \quad (7.2)$$

$$L_j(z) = \log(1 + d_j z), \quad L_{\mathbf{u}}(z) = \sum_j u_j L_j(z), \quad C_{\ell}(\mathbf{u}) = [z^{\ell}] e^{-L_{\mathbf{u}}(z)}. \quad (7.3)$$

The logarithms are their analytic germs at $z = 0$. Explicitly,

$$C_{\ell}(\mathbf{u}) = \sum_{q_1 + \dots + q_m = \ell} \prod_{j=1}^m \frac{(-1)^{q_j} d_j^{q_j} (u_j)_{q_j}}{q_j!}. \quad (7.4)$$

For $\ell \geq 1$ this is a polynomial of degree at most ℓ and has zero constant term.

Theorem 7.1 (Finite polar numerator, normally convergent regular part). *The continuation of (7.1) is*

$$\mathcal{Z}_P(\mathbf{u}; \mathbf{a}) = \sum_{k=0}^D p_k \sum_{\ell \geq 0} C_{\ell}(\mathbf{u}) \zeta(U + \ell - k, b). \quad (7.5)$$

Near $\mathbf{u} = 0$ it has the exact form

$$\mathcal{Z}_P(\mathbf{u}; \mathbf{a}) = \frac{\mathcal{R}_P(\mathbf{u})}{U} + \mathcal{H}_P(\mathbf{u}), \quad \mathcal{R}_P = \sum_{k=0}^D p_k C_{k+1}. \quad (7.6)$$

Here $\mathcal{R}_P(0) = 0$, $\deg \mathcal{R}_P \leq D + 1$, and $\mathcal{H}_P$ is holomorphic. An explicit regular part is

$$\begin{aligned} \mathcal{H}_P(\mathbf{u}) = \sum_{k=0}^D p_k \left\{ \zeta(U - k, b) + \sum_{\substack{\ell \geq 1 \\ \ell \neq k+1}} C_\ell(\mathbf{u}) \zeta(U + \ell - k, b) \right. \\ \left. + C_{k+1}(\mathbf{u}) g_b(U) \right\}. \end{aligned} \quad (7.7)$$

Every infinite sum displayed here converges normally, with every fixed spectral derivative, away from its explicitly displayed poles.

Proof. Expand $(n + a_j)^{-u_j}$ as $(n + b)^{-u_j}$ times the binomial series in $d_j/(n + b)$, then multiply the expansions. Choose a radius ρ with $1/b < \rho < 1/\max |d_j|$; when all $d_j = 0$, any $\rho > 1/b$ will do. On compact $\mathbf{u}$ -sets the exponential in (7.3) is bounded on $|z| = \rho$, so $|C_\ell(\mathbf{u})| \leq M\rho^{-\ell}$. For sufficiently large ℓ , the Dirichlet series for $\zeta(U + \ell - k, b)$ and its fixed derivatives are bounded by a constant times $b^{-\ell}$, allowing harmless polynomial factors in ℓ . Since $b\rho > 1$, the resulting series converges geometrically. The same Cauchy argument on a slightly larger compact set handles $\mathbf{u}$ derivatives. This proves every interchange in the initial half-space and then proves (7.5) by continuation. For $|U| < 1/2$, the only Hurwitz poles occur when $\ell = k + 1$. Subtracting exactly their $1/U$ terms gives (7.6) and (7.7). $\square$

The germ need not be jointly holomorphic at zero: $\mathcal{R}_P$ need not be divisible by U . Its restriction to every ray $\mathbf{u} = s\mathbf{w}$ with $W := \sum_j w_j \neq 0$ is nevertheless holomorphic at $s = 0$, since $\mathcal{R}_P(0) = 0$. The auxiliary b can change $\mathcal{R}_P$ off $U = 0$ by a multiple of U , absorbed into $\mathcal{H}_P$. Its restriction to the polar hyperplane, and the tangent coefficient identities below, are intrinsic.

7.2 Polynomiality in normalized exponents and exact finite differences

For α satisfying $\sum_j \alpha_j = 1$, set

$$\mathfrak{F}_r(\alpha) = \left. \frac{d^r}{ds^r} \mathcal{Z}_P(s\alpha; \mathbf{a}) \right|_{s=0}, \quad r \geq 0. \quad (7.8)$$

Complex normalized exponents are allowed by continuation. Write $\mathcal{H}_{P,[q]}$ and $\mathcal{R}_{P,[q]}$ for the homogeneous Taylor parts.

Theorem 7.2 (Every normalized spectral jet is a polynomial). *One has*

$$\mathfrak{F}_r(\alpha) = r! \{ \mathcal{H}_{P,[r]}(\alpha) + \mathcal{R}_{P,[r+1]}(\alpha) \}, \quad (7.9)$$

$$\mathcal{R}_{P,[r+1]}(\alpha) = \frac{(-1)^{r+1}}{(r+1)!} \sum_{k=0}^D p_k [z^{k+1}] L_\alpha(z)^{r+1}. \quad (7.10)$$

Thus $\mathfrak{F}_r$ is a polynomial of degree at most $r + 1$ on the affine hyperplane $\sum \alpha_j = 1$. Its component of degree $r + 1$ is entirely local and vanishes when $r > D$.

Proof. Substitute $U = s$ in (7.6) and extract s^r . Homogeneous extraction in $e^{-L\alpha}$ gives (7.10). The degree assertions follow immediately. $\square$

For a function on the normalized hyperplane, put $\Delta_v f(\alpha) = f(\alpha + v) - f(\alpha)$ when $\sum_j v_j = 0$. The notation $[n^{-1}]f(n)$ means the coefficient of n^{-1} in the Laurent expansion at large n ; it is the negative of the complex-analytic residue at infinity when that residue applies.

Theorem 7.3 (Local polarization identity). *Let $v_1, \dots, v_{r+1}$ be tangent vectors with $\sum_j v_{ij} = 0$, and put $V_i(n) = \sum_j v_{ij} \log(n + a_j)$. Then*

$$\begin{aligned} \Delta_{v_1} \cdots \Delta_{v_{r+1}} \mathfrak{F}_r(\alpha) \\ = D_{v_1} \cdots D_{v_{r+1}} \mathfrak{F}_r(\alpha) = (-1)^{r+1} r! [n^{-1}] P(n) \prod_{i=1}^{r+1} V_i(n). \end{aligned} \quad (7.11)$$

All mixed finite differences of order $r + 2$ vanish. If $\deg P = r$ and p is its leading coefficient, the right side reduces to

$$(-1)^{r+1} r! p \prod_{i=1}^{r+1} \left(\sum_j v_{ij} a_j \right). \quad (7.12)$$

Proof. The $r + 1$ tangent differentiations kill $\mathcal{H}_{P,[r]}$ and polarize L_α^{r+1} , supplying a factor $(r + 1)!$ in (7.10). Since $\sum_j v_{ij} = 0$,

$$L_{v_i}((n + b)^{-1}) = \sum_j v_{ij} \log(n + a_j) = V_i(n) = O(n^{-1}).$$

The coefficient $\sum_k p_k [z^{k+1}] \prod_i L_{v_i}(z)$ is therefore $[n^{-1}] P(n) \prod_i V_i(n)$. Translation from $n + b$ to n preserves this coefficient. For a polynomial of degree at most $r + 1$, its mixed difference of order $r + 1$ equals its corresponding constant mixed derivative; for example integrate that derivative over the unit $(r + 1)$ -cube. This proves (7.11). Finally $V_i(n) = n^{-1} \sum_j v_{ij} a_j + O(n^{-2})$ proves (7.12). $\square$

Normalization cannot be omitted. For arbitrary w with $W \neq 0$, define $Z_w(s) = \mathcal{Z}_P(s w)$. Then

$$Z_w^{(r)}(0) = W^r \mathfrak{F}_r(w/W). \quad (7.13)$$

The degree theorem concerns factor exponents; it is not a polynomiality assertion in the shifts a_j themselves.

7.3 A full Hurwitz–Stieltjes formula for each jet

Put $A_{h,\ell} = [z^\ell] L_\alpha(z)^h$, with $A_{0,0} = 1$ and $A_{0,\ell} = 0$ for $\ell > 0$.

Corollary 7.4 (Normally convergent coefficient formula). *For every $r \geq 0$,*

$$\begin{aligned} \mathfrak{F}_r(\alpha) = \sum_{k=0}^D p_k \left\{ \zeta^{(r)}(-k, b) + \sum_{\substack{\ell \geq 1 \\ \ell \neq k+1}} \sum_{h=1}^r (-1)^h \binom{r}{h} A_{h,\ell} \zeta^{(r-h)}(\ell - k, b) \right. \\ \left. + \frac{(-1)^{r+1}}{r+1} A_{r+1,k+1} + (-1)^r \sum_{h=1}^r \binom{r}{h} A_{h,k+1} \gamma_{r-h}(b) \right\}. \end{aligned} \quad (7.14)$$

Empty sums vanish. The generalized Stieltjes constants occur linearly, with indices at most $r - 1$.

Proof. Multiply the expansion of $e^{-sL_\alpha(z)}$ by the Taylor expansion of each nonresonant Hurwitz factor in (7.5). For $\ell = k + 1$, multiply it by $s^{-1} + \sum_q (-1)^q \gamma_q(b) s^q / q!$. The coefficient of s^r receives a contribution from the exponential coefficient of order $r + 1$, which yields the third term in (7.14). Normal convergence was proved in Theorem 7.1. $\square$

This additional order- $r + 1$ coefficient must be retained before a spectral pole is canceled. Generalized harmonic numbers provide an alternative finite description of all binomial coefficients required:

$$\frac{(x)_q}{q!} = \frac{x}{q} \exp \left(\sum_{j \geq 1} \frac{(-1)^{j+1} H_{q-1}^{(j)}}{j} x^j \right), \quad q \geq 1. \quad (7.15)$$

This follows from $(x)_q / q! = (x/q) \prod_{j=1}^{q-1} (1 + x/j)$. Use it in (7.4) and truncate at the desired spectral order. Thus both the Stieltjes and generalized-harmonic content are explicit.

7.4 First derivative: the classical weighted anomaly and Gamma factors

Set

$$\zeta_P(s, a) = \sum_{n \geq 0} \frac{P(n)}{(n+a)^s} = \sum_{k=0}^D \frac{P^{(k)}(-a)}{k!} \zeta(s-k, a), \quad (7.16)$$

where continuation is understood, and define $\delta_w = \sum_j w_j \zeta'_P(0, a_j) - Z'_w(0)$. The first-derivative instance of the calculus recovers

$$\delta_w = \frac{1}{2W} \sum_{i < j} w_i w_j [n^{-1}] P(n) \log^2 \frac{n + a_i}{n + a_j}. \quad (7.17)$$

Indeed the nonlocal terms of $\mathfrak{F}_1$ are linear in the normalized weights and cancel against their interpolation at the vertices e_j . The local difference is

$$\frac{1}{2} \sum_k p_k [z^{k+1}] \left\{ \sum_j \alpha_j L_j^2 - \left(\sum_j \alpha_j L_j \right)^2 \right\} = \frac{1}{2} \sum_{i < j} \alpha_i \alpha_j \sum_k p_k [z^{k+1}] (L_i - L_j)^2.$$

Multiplication by W gives (7.17).

The sign convention is fixed by the determinant identity

$$e^{-Z'_w(0)} = e^{\delta_w} \prod_j e^{-w_j \zeta'_P(0, a_j)}. \quad (7.18)$$

For $P(n) = \sum_{d=0}^D q_d n^d$, the finite coefficient in (7.17) is

$$\sum_{d=1}^D q_d (-1)^{d+1} \sum_{\ell=1}^d \frac{(a_i^\ell - a_j^\ell)(a_i^{d+1-\ell} - a_j^{d+1-\ell})}{\ell(d+1-\ell)}. \quad (7.19)$$

For $P = 1$ it is zero; for $P = n, n^2, n^3$ it is, respectively,

$$\frac{(a_i - a_j)^2, \quad -(a_i - a_j)^2(a_i + a_j), \quad (a_i - a_j)^2(11a_i^2 + 14a_i a_j + 11a_j^2)}{12}.$$

For shifts $(1, 2, 3)$ and weights $(1, 2, 1)$, this gives $\delta = 1$ for $P = n$ and $\delta = -4$ for $P = n^2$.

With G denoting the Barnes G function normalized by $G(1) = 1$, $G(a+1) = \Gamma(a)G(a)$, the Hurwitz–Barnes identity gives, when $P(n) = n$,

$$\zeta'_P(0, a) = \zeta'(-1) - \log G(a+1) + \frac{a}{2} \log(2\pi). \quad (7.20)$$

For a direct normalization check, differentiate both sides using Hurwitz shift differentiation and the Barnes integral formula [7, Eq. 5.17.4]. Both derivatives are $a - 1/2 - a\psi(a)$, and their values at $a = 1$ agree. This proves the displayed relation on $(0, \infty)$. For two unit-weight factors the complete determinant is therefore

$$e^{-Z'_{(1,1)}(0)} = \frac{e^{(a-b)^2/4 - 2\zeta'(-1)} G(a+1) G(b+1)}{(2\pi)^{(a+b)/2}}. \quad (7.21)$$

Here $a, b > 0$ are the two shifts; b in this last formula need not be the expansion base used earlier. All logarithms of G are real on these positive arguments.

7.5 A six-term identity for second derivatives

Theorem 7.5 (Quadratic-weight Vandermonde evaluation). *Let P be a monic quadratic polynomial and, for $a, b > 0$, define*

$$\mathfrak{T}_P(a, b) = \frac{d^2}{ds^2} \sum_{n \geq 0} P(n) ((n+a)^2(n+b))^{-s} \Big|_{s=0}, \quad (7.22)$$

by meromorphic continuation of the initially convergent series. Then for every $a, b, c > 0$,

$$\begin{aligned} \mathfrak{T}_P(a, b) + \mathfrak{T}_P(b, c) + \mathfrak{T}_P(c, a) - \mathfrak{T}_P(b, a) - \mathfrak{T}_P(c, b) - \mathfrak{T}_P(a, c) \\ = \frac{2}{3}(a-b)(b-c)(c-a). \end{aligned} \quad (7.23)$$

For a quadratic of leading coefficient p , multiply the right side by p . For degree at most one, the right side is zero.

Proof. Use the three shifts (a, b, c) , take $\alpha_0 = (1, 1, 1)/3$, and take tangent vectors

$$\mathbf{v}_1 = (1, -1, 0)/3, \quad \mathbf{v}_2 = (0, 1, -1)/3, \quad \mathbf{v}_3 = (-1, 0, 1)/3.$$

Their sum is zero, so the empty and full subsets cancel in the third finite difference of $\mathfrak{F}_2$. Equation (7.12) gives

$$\sum_{i=1}^3 \{\mathfrak{F}_2(\alpha_0 + \mathbf{v}_i) - \mathfrak{F}_2(\alpha_0 - \mathbf{v}_i)\} = -\frac{2}{27}(a-b)(b-c)(c-a).$$

Each argument is a permutation of $(2/3, 1/3, 0)$. By (7.13) its value is the corresponding $\mathfrak{T}_P/9$. Reading the six arguments gives the negative of the left side of (7.23), divided by nine. This proves the formula and its sign. The lower-degree assertions follow from the coefficient $[n^{-1}]$ in (7.11). $\square$

For $(a, b, c) = (1, 2, 3)$ the value is $4/3$, independent of the lower two coefficients of P . This is a cancellation of all nonlocal second-derivative constants; applying only a first-derivative anomaly formula would not justify it.

7.6 An alternating Vandermonde identity with any number of factors

The six-term evaluation is the first nontrivial higher-derivative member of a general alternating family. Define

$$N = \binom{m}{2}, \quad \Delta(\mathbf{x}) = \prod_{1 \leq i < j \leq m} (x_j - x_i), \quad K_m = \prod_{k=0}^{m-1} k!,$$

and let $(\sigma \mathbf{w})_j = w_{\sigma(j)}$.

Theorem 7.6 (General spectral alternant). *For $m \geq 2$, shifts $a_j > 0$, and weights with $W = \sum_j w_j \neq 0$,*

$$\begin{aligned} \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) Z_{\sigma \mathbf{w}}^{(N-1)}(0) \\ = \frac{(-1)^N (N-1)!}{W K_m} \Delta(\mathbf{w}) [n^{-1}] P(n) \prod_{i < j} \log \frac{n + a_j}{n + a_i}. \end{aligned} \quad (7.24)$$

If $\deg P = N - 1$ with leading coefficient p , this is the complete algebraic evaluation

$$\boxed{\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) Z_{\sigma \mathbf{w}}^{(N-1)}(0) = \frac{(-1)^N (N-1)! p}{W K_m} \Delta(\mathbf{w}) \Delta(\mathbf{a}).} \quad (7.25)$$

For degree less than $N - 1$, the alternating sum is zero.

Proof. Every alternating polynomial in m variables is divisible by their Vandermonde polynomial, of degree N . Therefore alternation kills the degree-at-most- $N - 1$ term $\mathcal{H}_{P, [N-1]}$ in (7.9). The elementary alternant identity is

$$\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \left(\sum_j \alpha_{\sigma(j)} y_j \right)^N = \frac{N!}{K_m} \Delta(\boldsymbol{\alpha}) \Delta(\mathbf{y}). \quad (7.26)$$

To prove it, expand by the multinomial theorem. A monomial with repeated exponents disappears under alternation. The only distinct nonnegative exponent pattern of total degree N is a permutation of $0, 1, \dots, m-1$. The surviving determinant in the α_j and the corresponding determinant in the y_j give the two Vandermondes, with the positive orientation shown in (7.26) and multinomial factor $N!/K_m$.

Apply this identity to $y_j = L_j(z)$ in (7.10). The normalized alternating jet is

$$\frac{(-1)^N (N-1)!}{K_m} \Delta(\boldsymbol{\alpha}) \sum_k p_k [z^{k+1}] \prod_{i < j} (L_j(z) - L_i(z)).$$

At $z = (n+b)^{-1}$, every difference is $\log((n+a_j)/(n+a_i))$. The coefficient is therefore the one in (7.24). Use (7.13) and $\Delta(\mathbf{w}/W) = W^{-N} \Delta(\mathbf{w})$ to obtain the unnormalized factor $1/W$. Finally, the logarithmic product starts with $\Delta(\mathbf{a}) n^{-N}$, proving (7.25) and the lower-degree vanishing. $\square$

For $m = 3$ and $\mathbf{w} = (2, 1, 0)$, equation (7.25) is exactly (7.23). For $m = 4$, $N = 6$ and $(N-1)!/K_m = 10$: a signed sum of twenty-four fifth spectral derivatives equals $10p\Delta(\mathbf{w})\Delta(\mathbf{a})/W$. For example, with $\mathbf{w} = (3, 2, 1, 0)$, $\mathbf{a} = (1, 2, 3, 4)$ and any monic degree-five P , that sum is exactly 240. No individual fifth derivative is evaluated to obtain this algebraic cancellation.

7.7 Shift derivatives, polygammas, and normalized primitives

The single-factor spectral derivative has the exact shift derivative

$$\frac{d}{da} \zeta'_P(0, a) = -\text{FP}_{s=1} \zeta_P(s, a) = P(-a)\psi(a) - \sum_{k=1}^D \frac{P^{(k)}(-a)}{k!} \zeta(1-k, a). \quad (7.27)$$

To prove it, differentiate the convergent Dirichlet series to obtain $\partial_a \zeta_P(s, a) = -s \zeta_P(s+1, a)$. The latter zeta factor has residue $P(-a)$ at $s=0$. Multiplication by s before differentiation shows that its finite part, rather than an undefined value at the pole, is required. Since $\zeta(1-k, a) = -B_k(a)/k$, the result is a polynomial times digamma plus a polynomial. Repeated differentiation gives explicit polygamma identities; integration over $[a_0, a_1] \subset (0, \infty)$ gives the normalized primitive $\zeta'_P(0, a_1) - \zeta'_P(0, a_0)$.

Combining (7.27) with the finite anomaly polynomial gives all shift derivatives of the product determinant. For example, for $P = n$ and two unit weights,

$$\partial_a Z'_{(1,1)}(0) = \frac{a+b-1}{2} - a\psi(a).$$

7.8 Logarithmic multiplicities and higher-order poles

Fix the same expansion base b as in (7.2), and for an integer $h \geq 0$ consider the distinct spectrum

$$\mathcal{Z}_{P,h}(\mathbf{u}; \mathbf{a}; b) = \sum_{n \geq 0} P(n) \log^h(n+b) \prod_j (n+a_j)^{-u_j}. \quad (7.28)$$

Changing this b changes the logarithmic multiplicity itself. It must not be treated as a freely variable auxiliary base in this definition.

Theorem 7.7 (Higher-pole finite parts and their local identities). *Near zero,*

$$\mathcal{Z}_{P,h}(\mathbf{u}; \mathbf{a}; b) = \frac{h! \mathcal{R}_P(\mathbf{u})}{U^{h+1}} + \mathcal{H}_{P,h}(\mathbf{u}), \quad (7.29)$$

where $\mathcal{H}_{P,h}$ is holomorphic. For normalized $\boldsymbol{\alpha}$, define the Laurent jet

$$\mathfrak{J}_{r,h}(\boldsymbol{\alpha}) := r! [s^r] \mathcal{Z}_{P,h}(s\boldsymbol{\alpha}; \mathbf{a}; b), \quad r \geq 0. \quad (7.30)$$

It is a polynomial of degree at most $q := r + h + 1$, and

$$\mathfrak{J}_{r,h} = r! \{ \mathcal{H}_{P,h,[r]} + h! \mathcal{R}_{P,[r+h+1]} \}, \quad (7.31)$$

$$\Delta_{v_1} \cdots \Delta_{v_q} \mathfrak{J}_{r,h}(\boldsymbol{\alpha}) = h! (-1)^q r! [n^{-1}] P(n) \prod_{i=1}^q V_i(n) \quad (7.32)$$

for tangent vectors as before. All differences of order $q+1$ vanish.

Proof. The proof of (7.5) is unchanged, except that each Hurwitz factor is replaced by $(-1)^h \zeta^{(h)}$. At resonance,

$$(-1)^h \zeta^{(h)}(1+U, b) = \frac{h!}{U^{h+1}} + \sum_{j \geq 0} \frac{(-1)^j \gamma_{j+h}(b)}{j!} U^j.$$

Subtract the displayed pole to prove (7.29). Normal convergence persists after these fixed derivatives. Taking $[s^r]$ gives (7.31), and polarizing its degree- q part as in Theorem 7.3 proves (7.32). $\square$

If $r + h > D$, the residue contribution in (7.31) vanishes, sharpening the degree bound to r . In particular, if $h > D$, the constant finite part is independent of the factor shifts and normalized weights and equals $(-1)^h \sum_k p_k \zeta^{(h)}(-k, b)$, even though negative Laurent powers may remain.

There is also a higher-pole version of Theorem 7.6. Let $N = \binom{m}{2}$, $0 \leq h \leq N - 1$, and $r = N - h - 1$. The identical alternant proof applied to (7.31) gives

$$\begin{aligned} & \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) r! [s^r] \mathcal{Z}_{P,h}(s\sigma \mathbf{w}; \mathbf{a}; b) \\ &= \frac{h!(-1)^N r!}{W^{h+1} K_m} \Delta(\mathbf{w}) [n^{-1}] P(n) \prod_{i < j} \log \frac{n + a_j}{n + a_i}. \end{aligned} \quad (7.33)$$

For $\deg P = N - 1$ the final coefficient is $p\Delta(\mathbf{a})$. Thus the alternating algebraic evaluation persists for logarithmic multiplicities, with the pole order determining the required Laurent jet and the power of W .

For $h > 0$ the ray generally has a pole of order at most h . Consequently (7.30) is a Laurent coefficient, not an ordinary derivative of the unmodified spectrum at zero. With $\mu = \sum_j \alpha_j (a_j - b)$, two concrete constant terms are

$$\text{FP}_{s=0} \sum_{n \geq 0} n \log(n + b) \prod_j (n + a_j)^{-\alpha_j s} = -\zeta'(-1, b) + b\zeta'(0, b) + \frac{\mu^2}{2}, \quad (7.34)$$

$$\text{FP}_{s=0} \sum_{n \geq 0} n^2 \log^2(n + b) \prod_j (n + a_j)^{-\alpha_j s} = \zeta''(-2, b) - 2b\zeta''(-1, b) + b^2\zeta''(0, b) - \frac{\mu^3}{3}. \quad (7.35)$$

The positive-degree dependence of these finite parts is completely algebraic despite the higher spectral poles. These formulas resolve the concrete logarithmic-multiplicity extension of the local calculus; arbitrary nonpolynomial spectra require a separate continuation theorem.

8 Audit, integration status, and further research

8.1 Precise disposition of the source questions

The source baseline matters because the manuscript has already absorbed many earlier conjectures and corrections. The audit below concerns the relevant canonical sections and the five incoming reports at the pinned revision; it is not a claim that every line of the collective manuscript has been independently re-proved.

Source target	Result of this continuation	Remaining scope
Nested Harmonic Jets, question 7	Theorem 5.2 constructs every $F_R(-a - m; a)$. Theorems 5.1 and 5.3 make the coordinate construction finite and explicit.	The smallest arithmetic or differential-algebraic coordinate class is not proved.
Nested Harmonic Jets, question 4	Theorems 3.1–3.3 give every transverse ordered depth-two finite part, including curvature and reparametrization. Theorem 4.2 gives every fully symmetrized direction.	Individual ordered germs in higher depth and compatible renormalization remain separate problems.

Source target	Result of this continuation	Remaining scope
Nested Harmonic Jets, question 3	Independent perturbation directions have a complete finite coefficient algorithm after full labelled symmetrization.	This does not construct the general two-colour ordered product or every non-diagonal Bell model.
Canonical Gaussian S_6 and current S_8 candidates	Neither short arithmetic evaluation is proved here. The current source's conjectural labels must remain.	A relation outside the restricted fixed-weight product presentation is needed.
Inverse-colour parity	Section 6 supplies a two-shift disk formula, its derivative calculus, and complete complementary-diagonal reductions.	The underlying general parity and cyclotomic symmetry principles have prior proofs.
Polynomial spectral products	Section 7 proves an all-order normalized-weight calculus and local identities, including higher poles.	The first-derivative multiplicative anomaly retains its classical attribution.

Why the Gaussian candidates have not been promoted. The canonical manuscript contains an exact obstruction in its specified level-four product-row presentation: a separating functional vanishes on the Gaussian-supported rows but not on the mixed S_{2m} coordinate [1, chapters/05-1 evel4-rank.tex]. The already proved S_4 identity uses a larger vocabulary, so this is a proof-system obstruction, not an arithmetic independence theorem. The present shifted-gap reflection belongs to the established parity mechanism; simply specializing it does not supply the missing weight-seven or weight-nine certificate. The source also distinguishes its current frozen S_8 vector from a previously rejected vector. Neither a tiny residual interval containing zero nor equivalence of two candidate baskets proves the vanishing of the residual.

8.2 A confirmed external indexing correction

There is one concrete typographical correction in the external material. In Gürel [15, Theorem 1.1, printed p. 2], arXiv version 2 of 3 September 2025, the second exponential sum is indexed by ℓ but its denominator is printed as k . The original PDF, not only the HTML conversion, contains this index. The denominator should be ℓ .

Here is a direct derivation that fixes the correction. Under the regularizable-spectrum hypotheses of that theorem, write $R_\ell = \text{Res}_{s=\ell} \zeta_\Lambda(s)$ and $C_\ell = \text{FP}_{s=\ell} \zeta_\Lambda(s)$. Near $s = 0$,

$$\frac{(s)_\ell}{\ell!} = \frac{s}{\ell} + \frac{H_{\ell-1}s^2}{\ell} + O(s^3), \quad \zeta_\Lambda(\ell + s) = \frac{R_\ell}{s} + C_\ell + O(s).$$

The coefficient of s in their product is $(C_\ell + H_{\ell-1}R_\ell)/\ell$. Expanding the shifted spectral series and negating its derivative therefore gives the corrected form

$$D_\Lambda(z) = \exp \left\{ -\zeta'_\Lambda(0) - \sum_{\ell=1}^M \frac{C_\ell + H_{\ell-1}R_\ell}{\ell} z^\ell \right\} \Delta_\Lambda(z), \quad (8.1)$$

where $H_0 = 0$, and M and the canonical product Δ_Λ have the source's meaning. The convergent terms with $\ell > M$ form the logarithm of that canonical product. This is an indexing correction, not a refutation of the theorem.

8.3 Recommended clarifications for integration

No fresh substantive error was found in the relevant current repository proofs examined here. The following clarifications prevent errors when the new sections are integrated.

1. **Specify the regulator before a finite part.** Replace any informal reference to “the finite part at $(1, 1)$ ” by an explicit path or subtraction convention. Equations (3.8) and (3.9) are concrete transition formulas between these choices.
2. **Retain the labelled symmetry factor.** The fully symmetrized depth- d value has $d!$ copies on the diagonal. The source’s individual diagonal convention is recovered by dividing by $d!$, not by suppressing repeated permutations.
3. **Distinguish fixed and moving Gamma arguments.** The value $F_R(-a - m; a)$ holds z fixed while differentiating s . The path $z = -(a + m)^s$ gives zero identically. Their derivative constraints are compatible but are different statements.
4. **Multiply through a pole before differentiating.** The spectral jet of order r can require a coefficient of order $r + 1$; with pole order $h + 1$ it can require order $r + h + 1$. Equations (7.14) and (7.31) show the missing terms that a premature truncation would discard.
5. **Keep ordinary sums grouped and primitives normalized.** The terms of (2.4) are summed together. The combined right side of (5.7) is integrated together. The index-one primitive (6.10) subtracts a base point before summation. These are convergence statements in the theorems, not discretionary numerical conventions.
6. **Preserve scope and attribution.** Do not relabel general parity, the symmetric-sum identity, or the first multiplicative anomaly as newly proved conjectures. Do not turn the finite coordinate ceiling into a minimality theorem. Keep the current S_6 and S_8 status unchanged unless a separate exact certificate is supplied.

8.4 Reproducible checks and their evidential role

The accompanying verification programs implement independent representations where practical. The Gamma-zero calculations compare the coordinate formula with Cauchy extraction from the original spectral product. The shifted-gap checks compare bilateral reductions with a Mellin integral. The anomaly checks use a finite initial segment and an independently expanded Hurwitz tail before Cauchy extraction. Exact polynomial checks test the finite combinatorial identities and normalizations. The complete execution parameters, counts, residuals, and package versions are retained in the machine-readable results.

The main numerical checks include sixteen Gamma-zero evaluations through derivative order four, seven primitive-recursion checks, four general shifted-gap reflections and three half-diagonal cases, four first-derivative anomaly evaluations, and a six-term second-jet evaluation. Observed absolute residuals in these independently evaluated special-function checks are below 10^{-39} ; the recorded results give the individual values and precision. These are floating-point diagnostics, not rigorous interval certificates. Exact symbolic checks and the analytic proofs have separate roles.

The package contains modular L^AT_EX sources, the compiled article, verification code, recorded results, a source-provenance manifest, and integration notes with theorem labels. It uses no external files from the source repository to compile. The original source snapshot is identified by hashes rather than copied wholesale into the deliverable.

8.5 Further research questions with concrete first targets

- Q1. Ordered depth three, with independent exponents.** Construct an analogue of (3.2) for $Z_3(1+u, 1+v, 1+w; a)$, displaying the intersecting polar divisors and a normally convergent holomorphic remainder. A first deliverable is an explicit constant term along every transverse ray, with its symmetrization checked against (4.6). The comparison should use the existing Laurent-expansion methods of [9].
- Q2. A compatible subtraction algebra.** Determine the local counterterms that convert the directional constants of Sections 3–4 into a specified stuffle-preserving renormalization. The depth-two discrepancy $-\gamma_1(a)(c/d + d/c)$ is the first necessary compatibility test. Compare the result directly with [10]; path independence must be a proved property of the chosen prescription.
- Q3. Tangential and nonlinear approaches to nested poles.** Theorem 3.3 handles transverse analytic curves. Classify curves tangent to $u = 0$ or $u + v = 0$ by their vanishing orders and give the exact number of curve coefficients required for a finite part. Curves lying identically in a polar divisor need an independently specified restriction or subtraction operation.
- Q4. Smaller arithmetic classes for Gamma-zero coordinates.** For a specified positive rational x , decide whether $D(x)$ and the first higher coordinates $L_2(x), M_{2,2}(x)$ reduce to a stated field of Gamma, polygamma, zeta-derivative, Dirichlet- L -derivative, and algebraic-argument polylogarithm values. Fix the comparison field before searching. Proving irreducibility requires more than failure of symbolic simplification or an integer-relation search.
- Q5. Several isolated zero factors.** Develop the counterpart of Theorem 5.2 for a multicolour product in which two or more factors vanish at the same parameter. Determine how the multiplicity shifts the first nonzero normal derivative, and how mixed spectral derivatives are encoded by multivariate Bell polynomials. A two-colour periodic assignment is the first case connecting directly to the source’s question 3.
- Q6. The nonsymmetric Stieltjes sums.** Equation (2.8) evaluates the symmetric combination $S_{m,n} + S_{n,m}$. Determine useful exact formulas for the difference $S_{m,n} - S_{n,m}$, beginning with $(m, n) = (0, 1)$. Through (3.11), this is exactly a question about the nonsymmetric regular double-Hurwitz jet. Identify which part reduces to ordinary Stieltjes constants and which part needs a new normalized integral.
- Q7. Primitive bases for complementary-shift polylogarithms.** For $a \geq 2$, integrate (6.11) against dz/z in a specified classical or multiple-polylogarithm basis. The half-shift formula (6.21) fixes the first normalization. At rational shifts, track the root-of-unity alphabet explicitly and separate variable-argument identities from isolated constant reductions.
- Q8. Noninteger indices and global shift continuation.** Construct a contour or Mellin continuation of the two-shift reflection in noninteger exponents. An integer-index partial fraction with a finite summation range cannot be differentiated spectrally without such an extension. Separately continue the shifts beyond the safe strip, recording finite terms when lattice poles cross a contour.

- Q9. Spectral interpolation and representation components.** Classify finite identities between order- r jets sampled at rational normalized weights. Theorem 7.2 turns this into finite polynomial interpolation with a known top-degree correction. Beyond alternating sums, determine which permutation-representation components kill every nonlocal term and yield factorizations in Vandermonde or Schur polynomials.
- Q10. Arithmetic and nonpolynomial multiplicities.** Extend the all-order local calculus to quasi-polynomial or periodic weights by residue-class splitting, and then to spectra whose base zeta has several or higher-order poles. Section 7 already settles the concrete polynomial times logarithmic class. For new classes, state the continuation and normal-convergence hypotheses that justify every extracted coefficient.
- Q11. An exact certificate for S_6 , then the current S_8 .** Construct a weight-seven functional equation using a larger Cayley or iterated-integral vocabulary, reduce it to the frozen S_6 basket, and exhibit a finite symbolic certificate. The source’s separating functional is a useful test that a genuinely additional relation has entered the proof. Only after that mechanism is understood should the weight-nine candidate be attacked by the same method. More digits alone do not resolve the structural obstruction.
- Q12. Formal interfaces for finite identities and analytic limits.** Formalize the subset telescoping law, partition Möbius inversion, Stirling rational-polylogarithm identity, Bell extraction, and tangent polarization as exact algebraic components. Then formalize normal convergence and Euler–Maclaurin remainder arguments as separate analytic lemmas. Interval implementations of the independent numerical checks would provide certified regression tests without being confused with the symbolic proofs.

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